

Development of Device for Non-Invasive Treatment of Alzheimer's Disease (AD) using Electric Field (EF)

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by

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GUIDE CERTIFICATE

This is to certify that the work presented in the report entitled "**Development of Device for Non-Invasive Treatment of Alzheimer's Disease (AD) using Electric Field (EF)**" by **Pooja (224159007)** represents original work under the guidance of **Prof. Harshal Bhalchandra Nemade and Dr. Vibin Ramakrishnan**. This study has not been submitted elsewhere for a degree.

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ABSTRACT

Alzheimer's disease (AD) is characterized by the accumulation of beta-amyloid peptide, forming plaques between nerve cells, and the formation of neurofibrillary tangles composed of tau protein within nerve cells. These abnormal protein clusters disrupt electrical signals and lead to a significant loss of connections among brain cells. This disruption results in inflammation and the death of brain cells, causing tissue shrinkage or atrophy. As the disease progresses, these plaques and tangles spread through the cortex, leading to a loss of memory, learning, and communication.

A proven effective strategy to stop the progression of AD is the promotion of proper folding and inhibition of aggregation of β -amyloid peptides. The application of electric and magnetic fields can alter the orientation and aggregation of such molecules in the brain. In-vitro analysis suggests that a uniform electric field of 150 V/cm is most effective in delaying the aggregation of beta-amyloid (Saikia et al., 2019). Neurons generate electric and magnetic fields due to the flow of electrical currents through their membranes and synaptic connections. Applied fields interact with these self-generated fields, helping to delay the progression of AD.

To advance this research for in-vivo application of electric fields in AD treatment, we developed a simple geometry of a mouse head, which includes the skin, skull, cerebrospinal fluid (CSF), cortex, and white matter. By applying voltage to this geometry, we achieved a uniform electric field of 150 V/cm at the cortex surface without spreading over the white matter.

Further optimization was conducted to reduce voltage requirements and improve the uniformity of the obtained electric field. This optimization involved varying the ratio of electrode width to the width of the gap. We created various iterations of the model with different width/gap ratios and compared them based on several parameters. A graph plotting uniformity against the width of electrode/width of gap ratio revealed that the optimal uniformity and minimum voltage requirement occurred when the width of the electrode was almost equal to the width of the gap.

Moving to an advanced 2D model, we incorporated additional details such as hair and a polyimide layer, the latter being the proposed material for flexible PCB manufacturing in the final device. Prior to this, we printed 3D models of the mouse skull and brain to ensure accurate dimensional understanding before advancing to the detailed 2D model. Through this comprehensive optimization and modeling, we determined that the maximum voltage requirement for the device could be limited to 25 volts.

Chapter 1

1. Introduction

Alzheimer's disease (AD) is a prevalent age-related protein misfolding disorder, posing a significant medical challenge due to its widespread occurrence, severity, associated costs, and the absence of targeted treatments (Guerriero et al., 2015). Representing 80% of reported dementia cases annually, AD is characterized by the misfolding, fibrillization, and deposition of β -amyloid proteins as senile plaques in the brain's gray matter. This process leads to neuronal dysfunction and death in crucial brain regions governing memory and cognition. Recent findings underscore the pivotal role of early oligomer intermediates, rather than mature fibrils, in severe neurodegeneration (S. Liu et al., 2014). The lack of a comprehensive understanding of disease progression is a major obstacle to developing effective therapies. Nevertheless, a key step in $A\beta$ fibrillogenesis involves an in vivo conformational transition from α (helical) to β conformation. $A\beta$ is generated through the proteolysis of the amyloid precursor protein (APP), initiated by β -secretase cleaving APP between Met671 and Asp672. Subsequent γ -secretase activity produces $A\beta_{1-40}$ and $A\beta_{1-42}$, which aggregate into oligomers or fibrils (Guo et al., 2022).

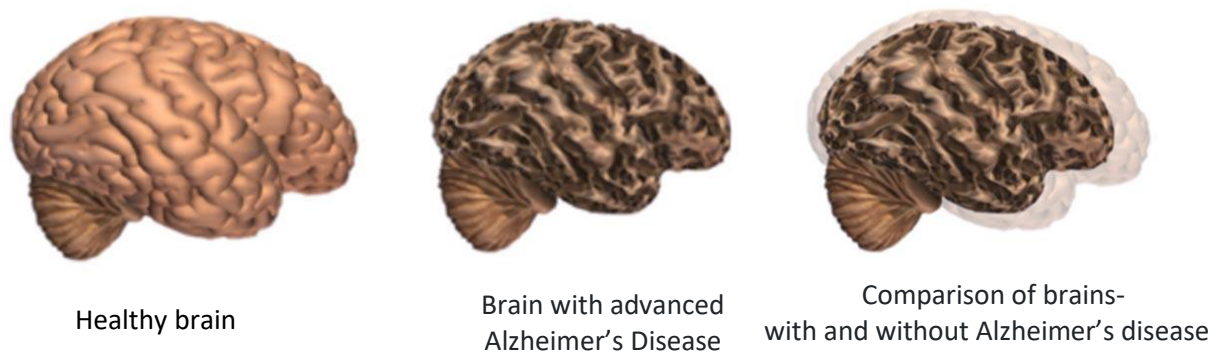


Figure 1: Comparison between sizes of healthy brain and brain with advanced AD

1.1. Stages in progression of AD:

1.1.1. Earliest Alzheimer's:

During the initial stages of Alzheimer's disease (AD), a covert and insidious molecular cascade unfolds, preceding the manifestation of clinically detectable symptoms by a considerable duration, often surpassing two decades before a definitive diagnosis can be established using current diagnostic modalities. This early pathological progression is characterized by the aberrant aggregation of amyloid-beta ($A\beta$) peptides into senile plaques and the formation of neurofibrillary tangles within specific regions of the brain implicated in cognitive functions (L. F. Liu et al., 2023).

The deposition of $A\beta$ peptides as senile plaques in the gray matter represents a seminal event in the preclinical phase of AD. These plaques are composed of aggregated $A\beta$ peptides resulting from the proteolytic cleavage of the amyloid precursor protein (APP) (Kametani & Hasegawa, 2018). The aggregation propensity of $A\beta$ peptides gives rise to insoluble fibrillar structures that accumulate in extracellular spaces. Simultaneously, intracellular neurofibrillary tangles, primarily consisting of hyperphosphorylated tau protein, develop within neurons (Abedin et al., 2021). Tau protein undergoes abnormal post-translational modifications, leading to its aggregation into paired helical filaments, constituting the core of neurofibrillary tangles. These molecular aberrations, occurring in brain areas crucial for memory and cognitive processes, contribute to the eventual synaptic dysfunction, neuronal degeneration, and cognitive decline characteristic of AD (Alquezar et al., 2021).

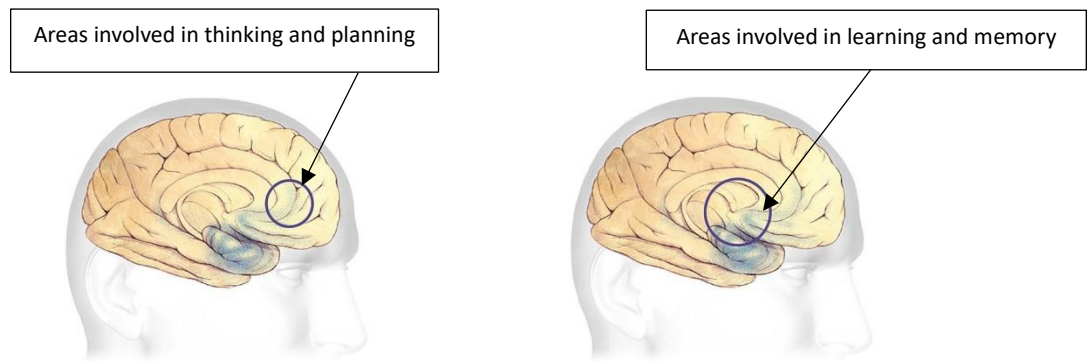


Figure 2 :Areas effected in earliest stage of AD

1.1.2. Mild to Moderate Alzheimer's:

During the transition from mild to moderate stages of Alzheimer's disease (AD), the pathological burden characterized by the accumulation of senile plaques and neurofibrillary tangles intensifies within brain regions crucial for memory, cognitive functions, and executive planning (Riquelme-Galindo et al., 2020). This augmented deposition imparts a discernible impact on individuals, leading to the emergence of noticeable memory and cognitive impairments that significantly impede occupational and social activities .

The enhanced aggregation of A β peptides into senile plaques and the progressive formation of neurofibrillary tangles, particularly rich in hyperphosphorylated tau protein, contribute to the exacerbation of cognitive deficits. Memory and thinking processes, integral to daily functioning, become markedly compromised. Individuals at this stage frequently encounter difficulties in memory recall, face challenges in organizing their thoughts, and experience a decline in problem-solving abilities. Manifestations extend beyond mere cognitive impairment to encompass disruptions in other domains, including confusion, impaired financial management, and difficulties in verbal expression (Hurtado et al., 2010).

Notably, a substantial number of individuals receive their initial diagnosis during the mild to moderate stages of AD due to the conspicuous nature of these cognitive and functional impairments. As the disease advances further, the pathological cascade extends its reach to additional brain regions involved in spatial orientation and language processing. Senile plaques and neurofibrillary tangles infiltrate areas crucial for the perception of one's body in relation to the surrounding environment and the intricate processes of speech comprehension and expression (Tyshchenko & Lukyanetz, 2020).

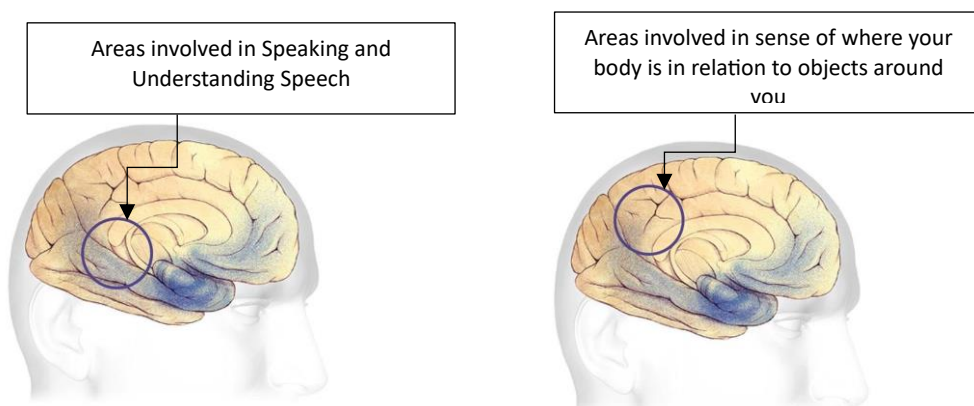


Figure 3:Areas effected in earliest stage of AD

1.1.3. Severe Alzheimer's Disease:

In the advanced stages of Alzheimer's disease (AD), a profound and extensive neuropathological impact becomes evident as a consequence of widespread cellular degeneration. The cortical regions of the brain, which are critical for various cognitive functions, experience severe damage, leading to a dramatic reduction in overall brain volume. This volumetric decline is primarily attributed to widespread neuronal cell death and synaptic loss, indicative of the advanced and irreversible nature of the disease (Weintraub et al., 2018).

At this stage, individuals with severe Alzheimer's disease undergo a profound deterioration in cognitive and functional capacities. Communication abilities are profoundly compromised, with individuals losing the capacity to articulate coherent thoughts and comprehend verbal or written language. Recognition of family members and loved ones becomes increasingly challenging, reflecting the extensive neural damage affecting memory and associative processes (Weintraub et al., 2018).

Furthermore, the progressive loss of cognitive and executive functions leads to a profound decline in the ability to perform basic activities of daily living. Individuals are no longer capable of independently caring for themselves, necessitating extensive assistance for tasks such as dressing, feeding, and personal hygiene. The severe impairment of motor skills further contributes to the overall decline in functional independence (Semiatin & O'Connor, 2012).

As the disease reaches its advanced stages, individuals often become reliant on comprehensive care and support due to the profound cognitive and physical debilitation. The caregiving burden increases significantly as the affected individuals lose touch with their surroundings and personal history. The severe cognitive decline, coupled with the loss of self-awareness and the ability to express needs, underscores the immense challenges associated with managing and caring for individuals in the severe stages of Alzheimer's disease (Epperly et al., 2017).

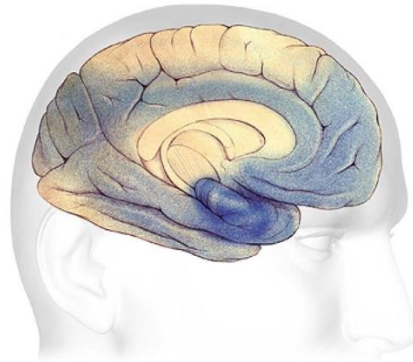


Figure 4: Severe Alzheimer's Disease (spread over cortex)

Scientific understanding of these advanced stages is crucial for the development of compassionate and effective care strategies aimed at enhancing the quality of life for both individuals affected by AD and their caregivers.

1.2. Proteins involved in AD pathogenicity, their functions, and structures:

1.2.1. Amyloid Precursor Protein (APP):

- **Normal Brain Function:** APP is a transmembrane protein with a role in neuronal growth, synapse formation, and repair processes in the normal brain. It is cleaved by enzymes in a non-amyloidogenic pathway, leading to the production of soluble fragments that have neuroprotective and neurotrophic functions (Epperly et al., 2017).
- **In AD:** Aberrant processing of APP in the amyloidogenic pathway results in the formation of beta-amyloid (A β) peptides. These A β peptides aggregate to form plaques, a hallmark pathology of Alzheimer's disease (AD). The accumulation of A β plaques is associated with

neurotoxicity, synaptic dysfunction, and neuroinflammation, contributing to AD pathogenesis (Schmidt et al., 2007).



Figure 5: Amyloid Precursor Protein (APP) [source: 2LLM]

1.2.2. Beta-Amyloid (A β):

- **Normal Brain Function:** The exact physiological role of A β in the normal brain is not fully understood. Some evidence suggests it may play a role in synaptic plasticity and neuronal activity regulation (Rogers et al., 1992).
- **In AD:** A β peptides, particularly A β (1–42), form insoluble aggregates and plaques in AD brains. These plaques disrupt neuronal communication, induce oxidative stress, and contribute to neurodegeneration. A β is implicated in the formation of toxic oligomers and fibrils, leading to synaptic dysfunction and neuronal death (Piccini et al., 2005).

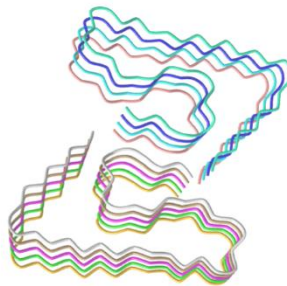


Figure 6: Beta-Amyloid (A β) [Source: 5OQV]

1.2.3. Tau Protein:

- **Normal Brain Function:** Tau is a microtubule-associated protein crucial for stabilizing microtubules and facilitating axonal transport in neurons. In its normal state, tau promotes the structural integrity of neurons (Lin, 2023).
- **In AD:** Abnormal hyperphosphorylation of tau disrupts its normal function, leading to the aggregation of tau proteins into neurofibrillary tangles. Neurofibrillary tangles are a key pathological feature of AD and are associated with neuronal dysfunction and cell death. Tau pathology correlates with cognitive decline in AD (Kolarova et al., 2012).



Figure 7: Tau Protein [Source: 2MZ7]

1.2.4. Apolipoprotein E (ApoE):

- **Normal Brain Function:** ApoE is involved in lipid transport and synaptic maintenance. It plays a role in neuronal repair and supports the overall health of neurons (Yamazaki et al., 2019).
- **In AD:** The APOE gene has different isoforms, and the APOE ϵ 4 allele is a major genetic risk factor for late-onset AD. Individuals with the ϵ 4 allele have an increased risk of developing AD, possibly due to its involvement in A β clearance and neuroinflammation (C. C. Liu et al., 2013).



Figure 8: Apolipoprotein E (ApoE) [Source: 1B68]

Table 1: Proteins involved in AD and their dysfunctions

Protein Name	PDB ID	Function in Normal Brain	Dysfunction in AD Brain	Ref.
Amyloid Precursor Protein (APP)	2LLM	Neuronal growth, synapse formation, and repair processes	Aberrant processing leads to A β formation, contributing to plaque formation, synaptic dysfunction, and neurotoxicity.	(Müller & Zheng, 2012)
Beta-Amyloid (A β)	5OQV	Exact physiological role not fully understood; may play a role in synaptic plasticity.	Aggregation into plaques, inducing neurotoxicity, synaptic dysfunction, and contributing to neurodegeneration.	(Haass & Selkoe, 2022)
Tau Protein	2MZ7	Stabilizes microtubules, facilitating axonal transport.	Hyperphosphorylation leads to tau aggregation into neurofibrillary tangles, contributing to synaptic dysfunction and neuronal death.	(Lin, 2023)
Apolipoprotein E (ApoE)	1B68	Involved in lipid transport and synaptic maintenance.	APOE ϵ 4 allele is a genetic risk factor, possibly impacting A β clearance and neuroinflammation in AD.	(Yamazaki et al., 2019)

1.3. Plaque and tangle formation:

Molecular characteristics of Alzheimer's disease

Amyloid plaques and neurofibrillary tangles that accumulate in an Alzheimer's brain consist of amyloid fibrils with different components but similar tertiary protein structures.

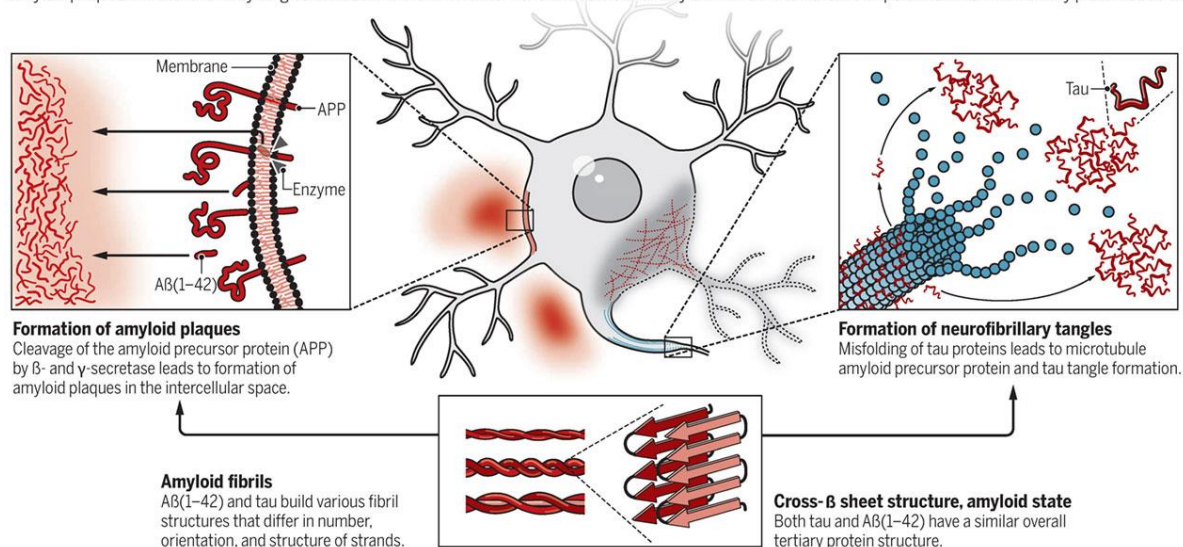


Figure 9: Plaques surround the neuron and tangles in a neuron

During the progression of AD, fibril accumulation undergoes exponential expansion due to the fragmentation of existing fibrils and subsequent prion-like seeding events.

Table 2: Difference between Neurofibrillary Tangles and Amyloid Plaques

Characteristic	Neurofibrillary Tangles	Amyloid Plaques
Composition	Aggregates of hyperphosphorylated tau protein.	Aggregates of beta-amyloid ($A\beta$) peptides.
Location	Found inside neurons.	Found outside neurons, in the extracellular space.
Microscopic Appearance	Twisted filaments forming intracellular tangles.	Dense, insoluble deposits of $A\beta$ peptides.
Primary Protein Component	Tau protein.	Beta-amyloid ($A\beta$) peptides derived from APP.
Association with Neuronal Death	Correlated with neuron dysfunction and death.	Correlated with synaptic dysfunction and neurotoxicity.
Effect on Synapses	Contributes to synaptic dysfunction and loss.	Induces synaptic dysfunction and may lead to loss.
Behavioral Correlation	Correlates with cognitive decline and memory loss.	Correlates with cognitive impairment and behavioral changes.
Temporal Formation	Occurs later in the course of AD.	Can develop earlier in AD progression.
Spread Through the Brain	Spreads through a stereotypic pattern in AD brains.	Also exhibits a stereotypic spread pattern in the brain.

Association with Prion-Like Properties	Can acquire prion-like properties.	A β peptides can also exhibit prion-like behavior.
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1.4. Therapeutic Strategies for Alzheimer's Disease:

Efforts to develop effective therapeutic strategies for Alzheimer's disease (AD) have been concentrated on multifaceted approaches targeting the molecular processes involved in the pathogenesis of the disease. One key strategy involves inhibiting the production and accumulation of amyloid-beta (A β), a hallmark of AD pathology. This includes the design of inhibitors aimed at blocking the expression of Amyloid Precursor Protein (APP), an upstream molecule in the A β production pathway. Additionally, interventions to prevent the proteolytic cleavage of APP into A β and strategies to clear A β fibrillar aggregates from the brain have been explored. Immunotherapy, both active and passive, has shown promise in clearing A β , but challenges such as brain hemorrhage and meningoencephalitis have been associated with this approach. Another avenue involves the development of β -sheet breaker peptides based on the stereochemical structure of A β 1-42. These peptides aim to destabilize the amyloidogenic A β conformer, hindering amyloid formation. However, peptide-based drugs face challenges related to metabolic instability and poor permeability through the blood-brain barrier (BBB) (J. Liu & Li, 2019).

In the realm of noninvasive therapeutic approaches, various methods have been investigated to perturb protein aggregation. Modified physiological conditions, such as temperature, pH, and solvents, have been employed to influence protein aggregation states. Beyond conventional approaches, external physical stimuli, such as microwave radiation, ultrasound, electric fields (EF), and magnetic fields (MF), have garnered attention for their potential impact on protein aggregation. Electric fields have been shown to induce structural changes in proteins and peptides, aligning molecular dipoles and influencing conformational transitions. Similarly, magnetic fields can exert torque on peptide structures, leading to ordered arrangements. Recent studies have delved into the effects of external electric and magnetic fields on the aggregation and self-assembly of amyloidogenic peptides.

In the upcoming chapter, we will conduct a comprehensive evaluation of the effectiveness and challenges pertaining to the treatment strategies discussed above. Our focus will be on an in-depth exploration of neurotechnological devices, delving into their current clinical statuses. Furthermore, we will intricately examine the impact of Electric Fields (EF) and Magnetic Fields (MF) on amyloid plaques. This section promises a thorough investigation into the existing landscape of neurotechnological interventions and an extensive exploration of the nuanced effects of EF and MF on the crucial aspect of plaque formation (Bakrim et al., 2022).

Chapter 2

2. Literature Review

Alzheimer's disease (AD) manifests as continuous decline in cognition and memory loss, marked by three key pathological markers: (i) β -amyloid ($A\beta$) plaques in extracellular place, generally composed of the $A\beta$ peptide which are derived from the amyloid precursor protein (APP) through β - and γ -secretase cleavage, (ii) neurofibrillary tangles (NFTs) inside the cells containing hyperphosphorylated tau (p-tau), and (iii) neuroinflammation due to microglial activation and astrogliosis. The accumulation of $A\beta$ disrupts signaling in cells, leading to NFTs, nutrient deprivation, cell death, and neuroinflammation. Traditionally, FDA-approved clinical treatments strategies for AD fall into two classes: N-methyl-D-aspartate (NMDA) receptor antagonists and Cholinesterase inhibitors. Despite offering symptomatic relief, these drugs have limited efficacy as disease-modifying treatments. Research has identified potential targets, including APP, γ -secretase (GSM), β -secretase (BACE1), Apolipoprotein-E, and recently, the innate immune system of brain targeting neuroinflammation and microglial clearance (Zhou et al., 2020).

The majority of Alzheimer's disease (AD) clinical trials have focused on targeting β -amyloid ($A\beta$), either by inhibiting its generation using BACE1 or γ -secretase inhibitors, or by directly addressing $A\beta$ aggregation and removal through active and passive immunization with anti- $A\beta$ antibodies. $A\beta$ immunotherapy, exemplified by aducanumab, utilizes antibodies directed against $A\beta$ fibrils and oligomers in the brain. Although aducanumab was FDA-approved in year 2021, clinical trial data analysis revealed no significant effects on symptoms, prompting the halt of the third phase clinical trial program of aducanumab. Subsequent amendments to the trial, involving higher doses administered over an extended period, reported a notable but modest slowing of decline in cognition in early-stage of AD. These findings underscore the potential efficacy of treatment at the initial stages of AD or even pre-symptomatically (Sciacca et al., 2018).

Numerous prior clinical trials targeting $A\beta$ faced setbacks, often because of safety concerns such as Amyloid-Related Imaging Abnormalities vasogenic edemas (ARIA-E) or hemorrhages (ARIA-H). Examples include $A\beta$ immunotherapies such as Crenezumab, Bapineuzumab, and Gantenerumab. Concurrently, γ -secretase inhibitors Semagacestat and Avagacestat increased symptoms after treatment. An alternative and more promising approach involves mitigating the generation of longer $A\beta$ isoforms, particularly $A\beta_{42}$, which are more prone to formation amyloid. This can be obtained by altering the docking site of β -secretase with γ -secretase modulators (GSM). Importantly, GSMs are designed not to resist γ -secretase cleavage of other substrates, such as NOTCH (Pozdnyakov et al., 2013).

The intricacy of Alzheimer's disease (AD) indicates that an effective treatment strategy may involve a combination of targets at different stages of disease. AD pathology initiates decades before the appearance of clinical symptoms, emphasizing the need for an early detection and intervention strategy. Neuropathogenicity of AD can be categorized into three scales: (i) microscale involving mutations in genes and molecular aberrations, (ii) physiological scale encompassing neuroimmunity, homeostasis, and dysfunctions in clearance, and (iii) at global scale involving dysfunction in circuit. While the conventional pharmacological approach usually focuses on a single scale, a combination of strategies targeting each of these areas may prove essential. Although this approach introduces added complexity in understanding the systemic, pathological, and molecular changes in AD, it could represent one of the most effective pathways to control and halt the disease progression (Thambisetty, 2017).

2.1. Exploring options beyond targeting $A\beta$:

Recent years have witnessed a growing number of clinical trials focused on tau, a microtubule-associated protein. Tau, integral to the neuronal cytoskeleton, plays roles in intracellular trafficking and microtubule stabilization. Tauopathy, characterized by tau pathology, spans various neurological

diseases, including progressive supranuclear palsy, traumatic brain injury, stroke, frontotemporal lobe dementia, Pick's disease, and AD. Notably, in AD, Tau and A β dysfunction are intricately linked, with evidence indicating tau as a downstream effector of A β , while other studies propose a feedback loop relationship. Recent drug targets for tau focus on regulating post-translational modifications, such as hyperphosphorylation, acetylation, and glycosylation, as well as addressing aggregation. These modifications can enhance clearance and impede tau accumulation, for instance, through the use of phosphodiesterase inhibitors. Additionally, approaches to directly reduce tau expression using small interfering RNA or antisense oligonucleotides, as well as the development of active and passive antibodies against tau, are actively pursued (Q. Q. Li et al., 2014).

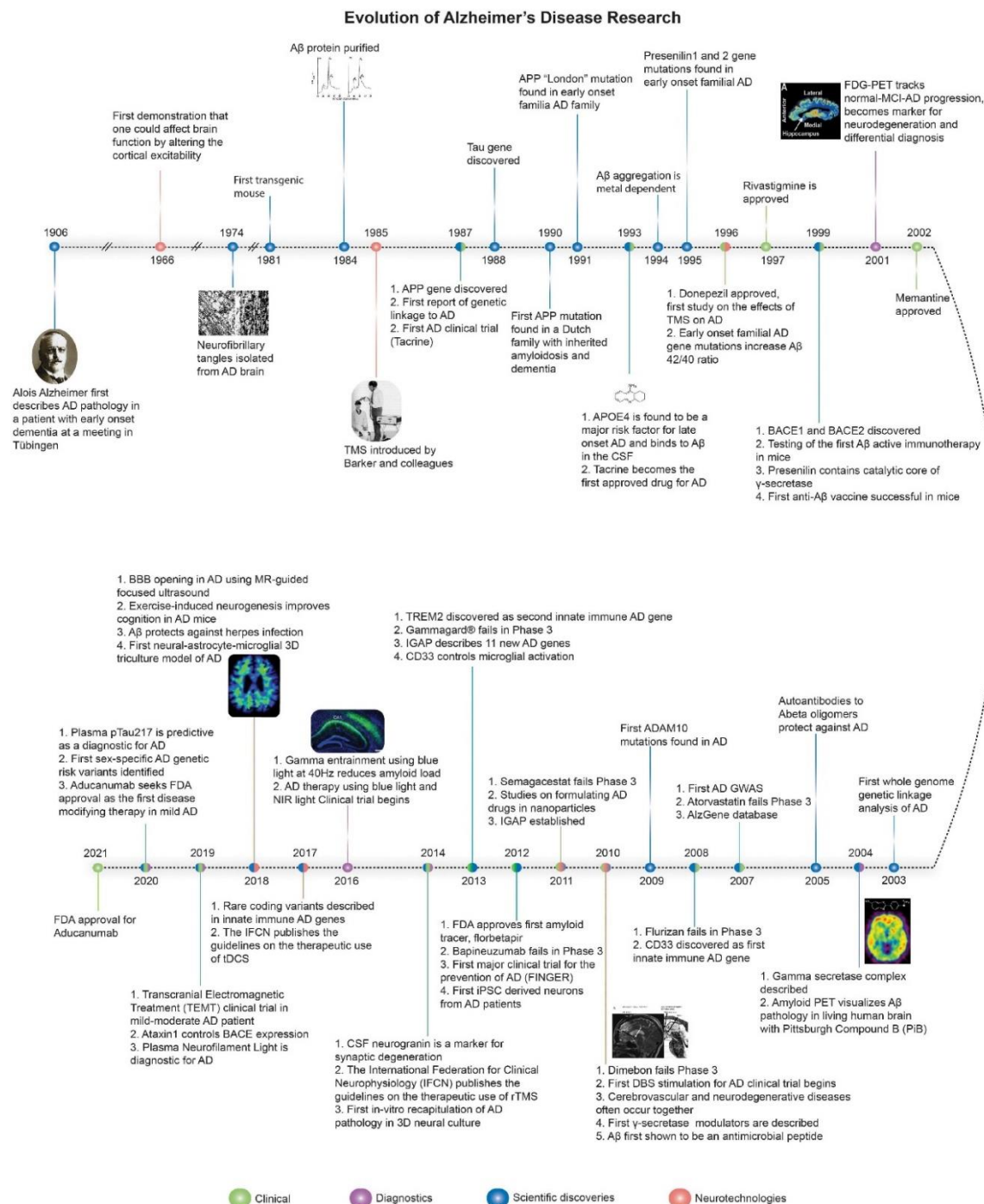


Figure 10: Timeline of major researches in AD

Astrocytes and reactive microglia envelop both A β plaques and neurons succumbing to tangles, leading to disrupted homeostasis and damage in neurons. This, in turn, can fuel chronic inflammation, creating a detrimental feedback loop. Consequently, molecules capable of modulating and regulating the neuroinflammatory process present novel therapeutic opportunities for AD treatment. One such molecule under investigation for targeting neuroinflammation in AD management is curcumin. Known for its anti-inflammatory properties, curcumin has been employed as a sensitive fluorochrome in conjunction with various imaging modalities. Its potential to suppress inflammatory cascades and attenuate AD progression, possibly by reducing A β levels and inhibiting A β aggregation, makes it a promising candidate. However, rigorous testing and further investigation are essential before these potential drugs and targets can be implemented in clinical practice (Serrano-Pozo et al., 2013).

Summarizing the progress in AD treatment over the past decades reveals a formidable challenge in developing effective therapeutics. The failure of a significant proportion of amyloid-based drug therapies has cast doubt on the amyloid cascade hypothesis. Notably, most failed phase three trial programs involved patients with mild-to-moderate symptomatic AD. Given that AD pathology accumulates years before symptom onset, addressing neuroinflammation as the primary driver of synaptic and neuronal loss in the preclinical phase may offer a more promising approach. For symptomatic patients, where neuroinflammation plays a central role in the damage, it becomes a key target. Successful AD treatment and prevention will likely necessitate a combination of amyloid as well as non-amyloid strategies, encompassing GSM, immunotherapeutics, anti-tau pathology agents, anti-neuroinflammation, APOE-based strategies, and even anti-microbial approaches.

Developing a single drug for AD is an expensive and lengthy process, averaging about \$5.7 billion and often taking over a decade from preclinical stages to potential FDA approval. Challenges abound, such as the growing competition to enroll early-stage AD patients in trials, difficulties in measuring functional assessments and cognitive tasks to achieve reliable data, and the inability to test human functional assessments in AD animal models. Moreover, differences between human and rat biology pose obstacles in translating scientific findings from the lab to clinical applications. Hence, leveraging recently evolved in vitro models like embryonic stem cell-derived cultures and three-dimensional (3D) human induced pluripotent stem cell (iPSC) becomes crucial to replicate the intricate human brain environment and expedite drug discovery. These models offer significant value in comprehending the underlying mechanisms in AD pathogenesis (Roman et al., 2023).

In the upcoming sections, we develop into the realm of emerging non-pharmacological neurotechnologies designed to tackle AD pathologies, employing light and electrical stimulation techniques. We'll emphasize the burgeoning array of non-invasive neuromodulation methods now being utilized as treatment options for AD. Specifically, our focus will be on exploring how electric field stimulation plays a role in alleviating AD-related pathogenesis. We will equally focus on the effect of electric fields on the in-vitro plaques of beta-amyloid. Our objective is to assess how these evolving neurotechnologies, alongside non-invasive brain stimulation methods using diverse approaches, could offer enhanced precision in dealing with the intricate clinical aspects of AD. Ultimately, our aim is to aid in devising more effective strategies for the treatment and prevention of this debilitating disease.

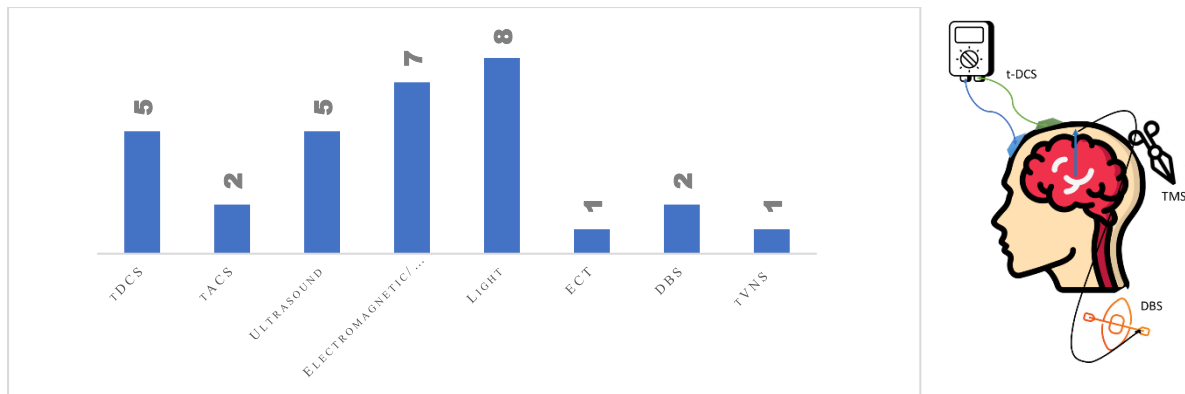


Figure 11: Neurotechnological device in clinical trials of AD treatment

2.2. Light stimulation and optogenetics in treatment of Alzheimer's Disease

Optogenetics has emerged as a groundbreaking tool in neuroscience, revolutionizing not only methodologies but also the conceptual framework guiding research in the field. Initially discovered in the archaeon *Halobacterium salinarum* in 1971, bacteriorhodopsin intrigued biologists due to its light-responsive proton and chloride transport functions. Years later, these light-gated opsins were adapted for neuronal control, giving rise to optogenetics. By employing light-sensitive ion channels from bacteria like channelrhodopsin, halorhodopsin, and archaerhodopsin, optogenetics allows the precise manipulation of cell membrane potentials using specific light wavelengths. These opsins can be introduced into cells via viral vectors containing the opsin gene, enabling the targeting of specific brain cells or regions and inducing behavioral changes (Kastanenka et al., 2019).

A newer method named "GENUS" (Gamma ENtrainment Using Sensory stimuli) has been developed, employing light flickering and 40 Hz sound waves to modulate brain cell activity. Research suggests that stimulating brain cells known as fast-spiking parvalbumin-positive (FS-PV)-interneurons at this frequency could reduce A β 40 and A β 42 levels in mice. However, the use of optical fiber implants for brain stimulation presents translational challenges. To address this, combining visual and audio stimulation at 40 Hz has shown similar effects on amyloid levels and demonstrated neuroprotective benefits (Chan et al., 2022).

Despite these intriguing findings, questions linger regarding how audiovisual stimuli traverse brain regions to affect A β load and neuroprotection. The transient nature of effects and the need for long-term treatments after repetitive GENUS sessions raise doubts. Moreover, these studies haven't fully explored the impact of GENUS on other types of cells involved in brain oscillations, such as astrocytes. Divergent responses across species also pose challenges. Initial trials applying LED light therapy at 40 Hz to AD patients showed no significant effects, potentially due to treatment duration and disease stage differences (Ismail et al., 2018). The effectiveness of these interventions in human patients might require prolonged treatments compared to those in mice.

Comparatively, optogenetics offers greater specificity in targeting different cell types, allowing precise spatial and temporal control. However, its use of invasive optical fibers and safety concerns related to viral gene delivery remain significant challenges. Some attempts have been made to create less invasive methods, like molecularly tailored nanoparticles for transcranial near-infrared (NIR) light stimulation. Nevertheless, issues with efficiency and potential neurotoxicity persist. Conversely, NIR light, used in photobiomodulation, may independently affect cognitive functions and holds promise as an AD treatment. Recently, a new light-sensitive opsin called SOUL demonstrated the ability to stimulate deeper brain regions without nanoparticles, showcasing similar specificity and control as optogenetics (Ning et al., 2022).

Designing neural probes/fibers for optogenetics presents challenges due to mechanical mismatches between commercial probes and soft brain tissue, potentially leading to tissue damage and scarring. Advances in materials development, like photoresponsive anti-A β agents and stimuli-responsive materials under NIR light, indicate potential therapeutic approaches. Yet, a non-invasive, precise tool for selective brain region stimulation with temporal accuracy remains a significant challenge (Cassano, 2021). Although these innovations pave the way for light-based brain stimulation's clinical potential, understanding their mechanisms and ongoing clinical trials are crucial to their advancement.

2.3. Ultrasound In treatment of Alziemer's disease:

Ultrasound, initially explored for tissue stimulation before its medical imaging application, has evolved significantly in neurological research. Harvey's early 20th-century experiments with high-frequency ultrasound waves altered the firing rate of amphibian cardiac muscles, while later, Fry et al. (1958) demonstrated its ability to temporarily suppress light-evoked potential in the visual cortex. Recent technological advancements have enabled more precise targeting at the millimeter scale using focused transducers, paving the way for low-intensity focused ultrasound as a non-invasive neurostimulation tool.

Studies on transcranial pulsed ultrasound for modulating neuronal activity in mice have highlighted its potential beyond the limitations of optogenetics. In primate and human studies, ultrasound has shown promise in enhancing sensory discrimination tasks and influencing visual and cognitive functions. Advancements in spatial resolution continue to be made, overcoming past challenges.

A unique aspect of ultrasound stimulation lies in its capability to bypass the blood-brain barrier (BBB). Magnetic resonance-guided focused ultrasound (MRIGFUS), coupled with microbubbles, has been explored to transiently open the BBB, potentially reducing plaque burden, promoting neuronal plasticity, and preventing spatial memory deficits in early-stage AD patients. Studies have reported tolerability and improved neuropsychological scores post-treatment, correlating with enhanced memory networks (M. Liu et al., 2018).

Moreover, MRIGFUS has been used to enhance the BBB permeability for delivering molecules, like TrKa agonists, directly improving cholinergic neurotransmission. This targeted approach addresses the challenge of spatial resolution in ultrasound stimulation, offering promising prospects for AD treatment by combining established chemical compounds with non-invasive neurotechnological approaches (Weber-Adrian et al., 2015).

Studies have demonstrated ultrasound's direct modulation of amyloid load, showing improved cognitive functions and reduced A β 42 levels in animal models. Researchers have also explored the potential of ultrasound-driven piezoelectric materials to destabilize A β fibrils, paving the way for potential biomedical applications, including drug delivery, neuromodulation, and cancer therapeutics. The coupling of ultrasound with nanobiomaterials holds promise in overcoming limitations in conventional treatments.

In summary, ultrasound emerges as a versatile non-invasive neurostimulation tool, showcasing potential in modulating neuronal activity, bypassing the BBB for drug delivery, and directly affecting amyloid load. These advancements hold promise for future AD treatments by combining innovative technologies with biocompatible materials to address critical challenges in neurological disorders.

2.4. Nanovectors:

Nanoscale science has introduced a new level of precision in understanding molecular and atomic scales, leading to significant advancements in drug delivery and medical imaging. One of the key challenges in drug delivery has been the inability to specifically target drug distribution to the intended pathological site, resulting in systemic dispersion and the inability to concentrate drugs locally. This often leads to uncontrolled release profiles and undesired side effects associated with systemic drug

delivery. Nanovectors, broadly classified into lipid-based, non-lipid organic-based, and inorganic materials, have emerged as promising carriers for drug payloads. These nanovectors can be engineered to transport drugs and adorned with specific antibodies for active targeting to pathological sites (Shi, 2018).

Recent developments in nanovector-based drug delivery for Alzheimer's disease (AD) have explored methods to deliver therapeutic agents into the AD brain. Initial studies involved delivering existing drugs like rivastigmine using nanoparticles coated to enhance bioavailability and cross the blood-brain barrier. Advancements enabled direct genetic manipulation by delivering siRNA for BACE1 knockdown using targeted exosomes. Additionally, targeted nanoliposomes have emerged as potential delivery systems for AD, capable of carrying therapeutic molecules across the BBB by functionalizing the vectors with specific molecules facilitating transcytosis. However, the efficacy of these studies depends on the availability of effective drugs or known genetic targets (Croissant et al., 2018).

Moreover, nanovectors have been investigated as a direct therapeutic intervention, such as A β nanodepleters constructed from silica nanostructures, reducing amyloid load in AD mouse models by capturing monomeric A β and inhibiting aggregate formation. The safety profile of these nanodepleters after capturing A β remains a critical consideration.

When considering clinical applications, challenges arise concerning the chronic administration of nanomaterials for treating or preventing AD. Factors affecting biodistribution and half-life include material size, composition, surface chemistry, and ligand functionalization. Concerns regarding off-target effects and accumulation of nanomaterials in organs need to be addressed for translating nanomedicines into clinical settings. Surface modifications have been explored to mitigate these concerns by affecting cellular interactions and internalization. Each nanoformulation and its drug combination require comprehensive assessment for specific pharmacokinetics and potential toxicity (Jonas G. Croissant et al., 2017).

2.5. Magnetic Nanoparticles:

Magnetic nanoparticles have undergone significant advancements in medical applications since their introduction. These nanoparticles have been customized extensively to achieve precise targeting, spatial control, and biocompatibility in drug delivery, aiming to minimize side effects. Superparamagnetic iron oxide nanoparticles (SPIONs) are particularly promising due to their unique physical characteristics, including superparamagnetism, high saturation field, and reversible magnetic properties activated solely under an external magnetic field. Their adaptable surface chemistry allows for modifications catering to specific functions, enabling controlled drug delivery, cell separation, ion channel regulation, and cellular manipulation.

The versatile surface coating of SPIONs allows active targeting by attaching functional antibodies to specific sites of interest, proving beneficial in overcoming the blood-brain barrier (BBB) in neurological disorders. Modifying magnetic nanoparticles, as demonstrated in studies, has shown promise in passing through the intact BBB under the guidance of an external magnetic field. This adaptability in modifying magnetic nanoparticles, coupled with external control, has facilitated their use as a delivery vehicle for approved drugs, enhancing BBB penetration and improving drug distribution into the brain (Sharma & Pervushin, 2020).

The following sections will present an overview of how SPION properties have been utilized to develop more efficient diagnostic and therapeutic platforms for Alzheimer's disease (AD). Additionally, it will address the challenges associated with clinical translation and discuss the implications of these nanomaterials for the future of AD treatment.

2.6. Quantum Dots:

Quantum dots, a class of semiconductor particles with unique optical and electrical properties, are transitioning from nanotechnology to health sciences. Initially known for their role in labeling and sensing, their application has expanded to act as molecular actuators, particularly in disrupting molecular interactions within disease-related proteins. While their use as nanoprobes for Alzheimer's disease (AD) diagnosis has been explored, fewer studies have investigated their potential therapeutic applications.

Recently, researchers demonstrated that graphene quantum dots can prevent the fibrillation of α -synuclein, a protein implicated in Parkinson's disease, suggesting a possible application in halting AD-related pathologies. Other studies have attempted to disrupt amyloid protein interactions using quantum dots, effectively inhibiting A β fibril formation. This therapeutic approach shows promise in preventing the initial pathological development of AD. However, the *in vivo* use of quantum dots is limited due to their cytotoxicity. Strategies like encapsulating them within polymers or coating their surface with biocompatible materials have been suggested to mitigate this cytotoxicity issue. Despite these challenges, efforts are underway to address the safety concerns associated with quantum dots for potential therapeutic applications in AD treatment (Villalva et al., 2021).

2.7. Electric Current-based Neurotechnologies:

2.7.1. Deep Brain Stimulation:

In 46 A.D., Scribonius Largus, a Roman physician, pioneered the clinical application of electrical brain stimulation to treat headaches and gout, laying the foundation for our present-day utilization of similar principles in advanced forms of open-loop single-source electrical stimulation. The evolution of modern deep brain stimulation (DBS) began in 1987 when French neurosurgeon Alim-Louis Benabid reported the insertion of a microwire with four electrical contacts into a patient's thalamus, delivering high-frequency continuously-on electrical stimulation to alleviate tremors. This breakthrough led to explorations of open-loop high-frequency stimulation on various subcortical targets, revealing that thalamic stimulation was notably safer than bilateral thalamotomy, and similar findings emerged for pallidal stimulation.

The success and enhanced safety of DBS compared to lesion surgeries prompted its widespread adoption and extended its applications beyond movement disorders to include pain, major depression, Tourette syndrome, obesity, epilepsy, and dementia. Despite its success, the exact mechanisms of action behind DBS remain unclear. Initially perceived as a reversible lesion due to its similarity to surgical lesions on the same brain regions, it is now understood that DBS has multifaceted neuromodulatory effects. Studies indicate that high-frequency DBS induces local inhibitory effects in basal ganglia targets through various mechanisms like sustained neural membrane depolarization, sodium channel inactivation, and increased potassium currents. Additionally, DBS can modulate presynaptic terminals and activate axons and dendrites, leading to complex downstream effects and influencing activity in remote brain regions (Luo et al., 2021).

Recent studies exploring DBS as a potential therapy for Alzheimer's disease (AD) have shown promising results, albeit with limitations such as localized inflammation at the implant site and the need for precise targeting of brain regions. The surgical implantation of DBS devices carries inherent risks, and the resulting inflammation might contribute to chronic central nervous system (CNS) inflammation seen in AD pathogenesis (Watson et al., 2017). However, DBS implants have the potential to precisely target early AD stages and may influence various neural cell types, including immune cells, to mitigate AD pathologies. There's also a prospect for developing closed-loop adaptive systems that integrate sensing and stimulating capabilities within the implant, offering personalized disease management and

enabling monitoring through mobile devices. Such advancements hold potential for personalized treatment strategies in neurodegenerative and other neurological disorders.

2.7.2. Electric Implants:

The integration of electrical brain implants represents a burgeoning area in neurotherapeutics, focusing on enhancing neural recording capabilities and precision in the field of neurotechnologies. Encouraged by global initiatives, research aims to achieve substantial milestones in neural recording, harnessing advanced tools to capture neural activity from thousands of neurons with sub-millisecond temporal precision. This drive parallels the increase in neural recording capacity, reminiscent of the rise in transistors in integrated circuits, but current neural probe designs are limited by the use of single shaft microwires. These microwires fail to integrate effectively into neural tissue over extended periods, impeding a comprehensive understanding of long-term neural function necessary for functional neural prosthetics.

Electrode implantation into the brain induces a local neuroinflammatory and glial response, governed by various factors including electrode composition, shape, implantation procedure, and more. This elicits both acute and chronic responses, with acute effects typically subsiding within a week while the chronic response, emerging within the first month, poses a significant challenge to probe longevity. Chronic responses include neuroinflammatory reactions leading to the formation of a dense encapsulation layer around the probe, hampering its functionality by increasing impedance and reducing signal quality, among other adverse effects (Wellman et al., 2023).

Recent advancements in microfabrication techniques have birthed Utah and Michigan-type silicon multielectrode arrays, significantly enhancing simultaneous sampling capabilities. Neuropixel and Neuroseeker probes have exponentially increased sensing capacity, allowing for thousands of densely packed cortical sites to be monitored. However, despite these advancements, long-term stability of penetrating electrodes remains a significant challenge due to the persistent foreign body response.

Interdisciplinary research spanning material science, engineering, and neuroscience aims to mitigate the immunological response following electrode implantation. Strategies include biomimicry approaches using bioactive coatings or biocompatible materials, drug-eluting coatings, and the development of softening biomaterials inspired by nature. Tissue-like electronics, such as mesh electronics, have emerged as a fundamentally different approach, integrating components into a 3D ultra-flexible scaffold to minimize immune responses and improve neural integration (HajjHassan et al., 2008).

Breakthroughs in the development of biocompatible electrical neural probes hold promise for implantable diagnostic and treatment interventions for neurological disorders like Alzheimer's disease. These probes enable real-time diagnostics and drug delivery once challenges related to neuroinflammation are addressed. Advancements in bimodal neural probes that detect electrical and chemical signals concurrently while enabling drug delivery exemplify the potential of electrical implants as a novel modality for personalized diagnostics and treatments in neurological disorders, evolving toward closed-loop systems tailored to individual needs.

2.8. Field-based Neurotechnologies:

Unlike brain stimulation methods that necessitate invasive implants or delivery techniques, researchers in the past two decades have shifted focus towards non-invasive field based stimulation methodologies (HajjHassan et al., 2008).

2.8.1. Electric field in a neuron:

- ❑ Electric field in dendrites- Single EPSP magnitude is 0.2mV (Assumed). In a given point of time the distribution of the voltage along the dendrite is obtained by the cable equation with $V \neq f(t)$ and $\frac{\partial V}{\partial t} = 0$
- ❑ In space and time the dynamics of a single EPSP could be described by- $V_{(x,t)} = V_0 e^{-x/\lambda} \cdot e^{-t/\tau}$ space constant λ depends on the diameter of the dendrite but if we need rough approximation

we could consider that the dendrite has constant diameter of $1\mu\text{m}$ and we can use the calculated value for the space constant $\lambda \sim 353\mu\text{m}$ -

- ❑ $\vec{E} = -\nabla V = -\frac{dV(x,t)}{dx} = \frac{1}{\lambda} V_0 e^{-x/\lambda} \cdot e^{-t/\tau}$ The maximal electric field intensity for a single excitatory postsynaptic potential with magnitude of 0.2mV is $\sim 0.57\text{ V/m}$.
- ❑ Electric field in axons- Electric fields intensity is higher in axons because the space constant λ in axons is 1-2 orders of magnitude larger than the dendritic space constant. Voltage could vary from 70mV - 100mV (taking into account the overshoot of the action potential) so the electric intensity $\vec{E}$ could reach 10V/m .
- ❑ The magnetic and electric field strength in the soma are expected to be the same as in the dendrites, with a maximal magnetic strength of 10^{-10}T and maximal electric strength of 10V/m .

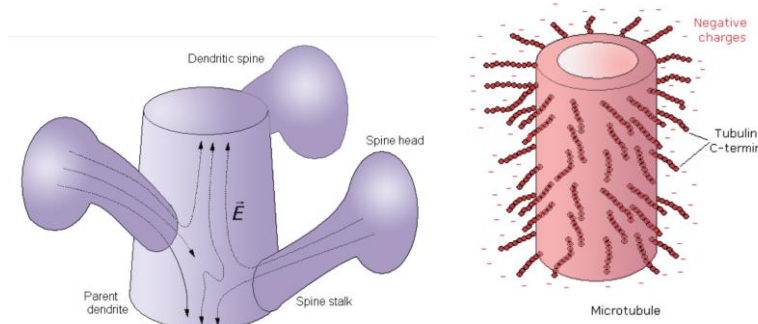


Figure 12: Electric fields in a neuron

2.8.2. Magnetic field in a neuron:

- ❑ The magnetic field is not strong enough to support the quantum Hall effect or provide information to microtubules. Each individual electric spike has a magnetic flux density three orders of magnitude smaller than the Earth's magnetic field, thus any information contained in the magnetic signal will be drowned out by background noise. On the other hand, the electric field operates on transmembrane voltage-gated ion channels that control the production of neuronal action potentials and contains biologically significant information.
- ❑ Magnetic field in dendrites-The currents inside dendrites are in 20pA - 100pA range for GABAergic synapses. $\oint \vec{H} \cdot d\vec{l} = i$
- ❑ The magnetic intensity for a contour Γ with length $l = \pi d$ that interweaves the whole current i_A . For dendrite with $d \sim 1\mu\text{m}$ we obtain- $H = \frac{100 \times 10^{-12}\text{A}}{3.14 \times 10^{-6}\text{m}} = 31.8 \times 10^{-6}\text{A} \cdot \text{m}^{-1}$
- ❑ Considering water and the microtubules form a system augmenting the magnetic strength known as ferrofluid then in the best-case with effective magnetic permeability $\mu_{\text{eff}} \approx 10$. Maximal magnitude of the magnetic induction $\vec{B}$ inside the neuronal projection: $\vec{B} = \mu_{\text{eff}} \mu_0 \vec{H} \approx 4 \times 10^{-10}\text{T}$
- ❑ The magnetic field in axons is stronger than that in dendrites due to the higher ion currents that flow through the axoplasm. The action potential can be described as two opposing current dipoles produced by a potential change of about 70mV . It takes the shape of a moving solitary wave. Peak currents range from $5\text{-}10\mu\text{A}$. Axons range in diameter from $1\text{-}25\mu\text{m}$ in humans- $B = \mu_{\text{eff}} \mu_0 \frac{i}{\pi d} = 1.6 \times 10^{-7}\text{T}$

Table 3: Effect of EMF exposure on symptoms related to AD pathophysiology in human

Types of EMF	Frequency/Level	Duration	Objectives	Main findings	Ref.
Extremely Weak (EW-EMF)	10–100 nW	5 weeks	To investigate in patients with a cognitive disability, episodic memory, and executive functions	Improvement in both immediate and delayed recalls of episodic memories	(Guerrier et al., 2015)
Extremely low (EL-EMF)	50 Hz	2 months	To study the relationship between ELF-EMF and the danger of AD	Increased risk of dementia only in individuals exposed to high levels of EMF	(Qiu et al., 2004)
Power frequency (PF-EMF)	50-1000 Hz	2-months	To study the relationship between exposure to PFEMF and the risk of cognitive impairment	Little support for the link between PF-EMF and cognitive impairment	(C. Y. Li et al., 2002)
Radio frequency (RM-EMF)	3 kHz–300 GHz	1 month	Test whether RF-EMF introduction alters memory through oxidative stress	No alterations in memory performance	(Bouji et al., 2020)

2.8.3. Transcranial magnetic stimulation (TMS):

Transcranial magnetic stimulation (TMS) and repetitive transcranial magnetic stimulation (rTMS) have been fundamental tools in cognitive neuroscience for years. By inducing changes in the brain's magnetic field through electrical currents in TMS coils, this technology can interrupt regular brain activity. TMS has been used to produce specific neuropsychological effects such as neglect syndrome and visual hallucinations, and explore functions like attention, perception, and learning. It also shows promise in treating movement disorders, epilepsy, depression, schizophrenia, and other psychiatric conditions. However, its application for understanding or treating neurological diseases, including Alzheimer's disease (AD), is still relatively new.

Studies employing TMS have shown increased motor cortex excitability in AD patients, likely due to impaired intracortical inhibition. Researchers have attempted to improve AD-related cognitive symptoms using TMS, such as anomia, comprehension, spatial learning, and memory. While clinical trials are exploring TMS as a potential AD therapy, its primary use has been observed in psychiatric disorders like depression. Recent findings even suggest that repetitive transcranial magnetic stimulation might preserve long-term memories by enhancing the brain's drainage system, introducing a potential novel approach for AD treatment through magnetic stimulation.

Precise targeting of magnetic force remains a significant challenge in TMS treatment. Researchers are investigating implanting magnetic coils to activate specific brain regions without requiring substantially higher magnetic fields. These technologies have found applications in neuronal stimulation, visual prostheses, brain-computer interfaces, and other areas. However, challenges persist in finding biocompatible materials that respond effectively to magnetic fields. Recent innovations, such as the m-Torquer developed by Lee et al. (2021), present promising steps by delivering piconewton-scale forces to neurons, enabling remote and consistent neuromodulation in freely moving subjects. These

advancements highlight the growing interest in magnetic neuromodulation in both fundamental research and potential clinical applications for conditions like neurodegeneration.

2.8.4. Transcranial current stimulation (tCS):

Transcranial Direct Current Stimulation (tDCS) has emerged as a non-invasive, safe, and cost-effective method for treating neurological disorders like Alzheimer's Disease (AD). It involves applying weak electrical currents to specific brain regions using scalp electrodes, modulating neuronal activity and potentially inducing synaptic plasticity.

While anodal tDCS increases cortical excitability, cathodal tDCS decreases it. Short-term effects are attributed to non-synaptic mechanisms, while long-term effects seem linked to NMDA-dependent mechanisms and synaptic plasticity. Despite minor adverse effects like tingling or mild headache, tDCS is generally considered safe and tolerable, except for patients with specific contraindications like metallic implants or a history of seizures.

Studies indicate positive effects of tDCS on AD and Parkinson's Disease (PD), particularly in the early stages. However, determining the most effective stimulation procedures remains a challenge. Researchers highlight the need for optimized protocols considering electrode placement, session number/timing, duration, and intensity. High-definition tDCS (HD-tDCS), employing smaller electrodes, offers precision in targeting brain regions and longer-lasting effects compared to conventional tDCS.

Exploring various stimulation sites, multisite techniques, or neurophysiological investigations using TMS/transcranial alternating current stimulation (tACS) and EEG activity could enhance our understanding of tDCS's efficacy in AD. Its application in the advanced stages of AD appears less effective, while it demonstrates promise in mild stages, often alongside cognitive or physical training.

With minimal side effects and reliable blinding in clinical settings, tDCS shows promise for improving cognition, psychopathological dimensions, and overall quality of life in AD and PD patients. However, larger randomized controlled trials (RCTs) are needed to replicate initial findings, measure diverse outcomes, and optimize tDCS protocols based on individual patient responses.

tACS utilizes low-intensity alternating electrical currents to influence brain activity by synchronizing specific cortical rhythms corresponding to the applied stimulation frequency.

In the context of Alzheimer's disease, employing tACS to entrain gamma oscillations (γ -tACS) holds promise in research due to its potential to re-establish typical brain function and enhance cognitive abilities. This potential is linked to the physiological mechanisms associated with gamma oscillation modulation, including its impact on neuroplasticity.

Technology/Therapy Name	Phenomenon Used	Pros	Cons
Transcranial Magnetic Stimulation (TMS)	Magnetic field induces electric currents in the brain	- Non-invasive, potential to modulate neuronal activity	- Limited evidence on long-term efficacy
Deep Brain Stimulation (DBS)	Uses electrical impulses to stimulate targeted brain areas	- Can alleviate symptoms, potential for personalized treatment	- Invasive, surgical procedure, potential risks
Transcranial Direct Current Stimulation (tDCS)	Applies low-intensity direct current to the scalp	- Non-invasive, considered safe and well-tolerated	- Limited understanding of optimal parameters, variable efficacy
Transcranial Alternating Current Stimulation (tACS)	Modulates brain activity using sinusoidal electrical current	- Potential for modulating brain oscillations, non-invasive	- Efficacy still being investigated, optimal parameters unclear

Brain Implants	Implantation of devices for electrical stimulation	- Precise targeting of brain regions, potential for personalized therapy	- Invasive, risk of complications, limited long-term studies
Quantum Dots	Semiconductor particles with unique optical/electrical properties	- Potential for molecular-level targeting, diverse applications	- Safety concerns, toxicity, and ethical considerations
Magnetic Nanoparticles	Nanoparticles for drug delivery, influenced by external magnets	- Controlled drug delivery, potential for targeting specific tissues	- Biocompatibility issues, potential toxicity, clearance concerns
Nano Vectors	Nano-scale carriers for drug delivery	- Targeted delivery of drugs, potential for crossing biological barriers	- Complex synthesis, potential toxicity, stability concerns
Immunotherapies	Utilizes immune system to target beta-amyloid or tau proteins	- Targets disease-specific proteins, potential disease-modifying effect	- Challenges in efficacy, safety, and immune response
Pharmacological-based Drugs	Chemical compounds targeting various aspects of AD pathology	- Wide range of targets, symptom management, some delay disease progress	- Limited efficacy, side effects, variable response among individuals
Ultrasound Stimulation	Uses focused ultrasound waves to modulate brain activity	- Non-invasive, potential for precise and targeted neuromodulation	- Safety concerns, optimal parameters and long-term effects unclear
Light Stimulation (Optogenetics/ Photobiomodulation)	Light-sensitive proteins or light to modulate neural activity	- Precise neural modulation, potential for targeted therapy	- Limited depth of penetration, complexity in delivery systems

Table 4: Comparison among treatment methods of AD

Chapter 3

3. Research Gap and Objectives

3.1. Research Gap

The landscape of available treatments for Alzheimer's disease (AD) presents a challenging scenario. Currently, about nine drugs have received FDA approval for AD treatment, with seven primarily focused on managing symptoms, while two employ immunotherapies to target beta-amyloid plaques. However, none of these medications directly address the underlying cause of AD or significantly alter its course at the molecular level. Emerging technologies such as neurotechnology, along with innovative approaches like quantum dots, nano-vectors, magnetic nanoparticles, and immunotherapies, are now being explored but they are way behind in their journey to be a potential treatment for AD. At present, there are about 33 devices undergoing clinical trials for AD treatment.

Among these devices, about eight are based on electric fields, while others utilize magnetic fields. Magnetic field-based devices often necessitate complex and expensive circuitry to generate a magnetic field of sufficient strength to alter brain circuitry, limiting their spatial precision. Among the electric field-based devices, promising outcomes have been observed with techniques like transcranial Direct Current Stimulation (tDCS) and transcranial Alternating Current Stimulation (tACS). However, the optimal parameters for these methods—such as frequency, magnitude, and field shape—remain uncertain. Presently, decisions regarding parameters for tACS, tDCS, and Electroconvulsive Therapy (ECT) are largely empirical, lacking specific studies elucidating the precise impact of diverse electric field configurations on the AD brain.

To treat Alzheimer's disease at molecular level, researchers are looking into application of electric field to remove or disintegrate plaques into smaller molecules which are less toxic for the neuron health. In-vitro studies say that low-intensity electric can inhibit plaque formation. Insights from this study could pave the way for the development of devices aimed at addressing the root cause of AD, potentially offering significant improvements in patient outcomes and quality of life.

3.2. Objectives

Based on the research gap, the following objectives were formulated such as:

- Design of prototype to find required magnitude of voltage range and shapes of electric field on rat brain
- Simulation of Geometry of mouse head in COMSOL Multiphysics
- 3D print of mouse skull and mouse brain for accurate measurement of dimensions
- Creation of tissue materials in COMSOL based on their electric properties
- Exploration of electrode combinations and applied voltages in COMSOL Multiphysics
- Optimization of applied voltage to obtain uniform electric field

Chapter 4

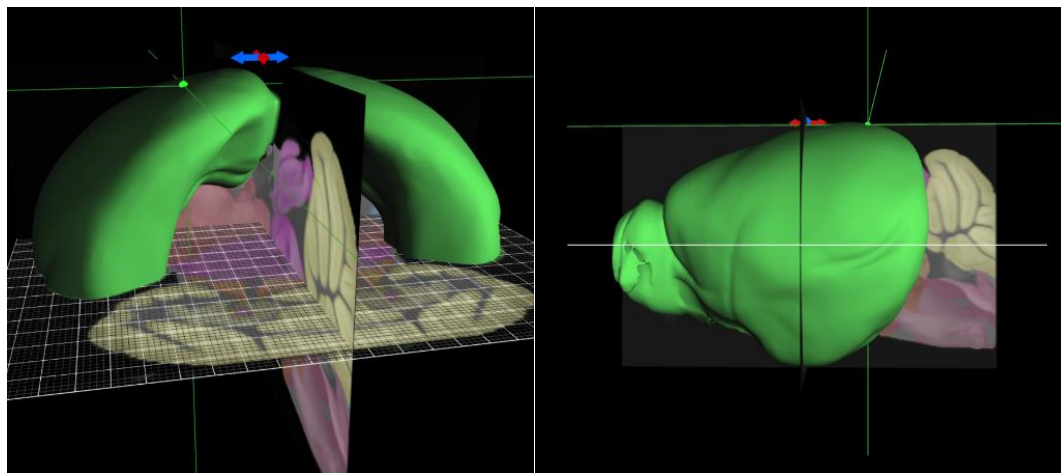
4. Methodology

4.1. Various phases and parts of project:

- Effective values of magnitudes of electric field, frequency and shape
- Uniform electric field profile over selected areas
- Voltage range required to deliver uniform electric field
- Architecture of electrodes for uniform electric field
- Power supply and wave shapes at different frequencies
- Integration of power supply components on a flexible PCB
- Packaging of electrodes, PCB and batteries
- Testing on rats

Effective values of magnitudes of electric field, frequency and shapes are decided on based of Jahnu Saikia et al.'s paper, "Electric Field Disruption of Amyloid Aggregation: Potential Non-invasive Therapy for Alzheimer's Disease". Paper explores the impact of electric and magnetic fields on the aggregation dynamics of A β 1-42 peptides. This study demonstrates a dose-dependent response in electric fields, leading to a discernible deceleration in the aggregation process. Lower electric field amplitudes exhibit an inhibitory effect that persists even after the field's removal, resulting in smaller and less toxic A β structures. This study emphasizes the potential for fine-tuning therapeutic applications based on specific amplitude parameters, offering a spectrum of possibilities for interventions. According to study most effective magnitude of an electric field is 150 V/cm. In literature review, it was reviewed that plaques and tangles are spread over gray matter (cortex), so effective portion that we consideration will be cortex.

This report includes development and simulation of geometry to obtain a uniform electric field spread over cortex of rat brain. In simulation, our goal is to obtain uniform electric field profile concentrated only on cortex of the mouse brain.



Back View

Side View

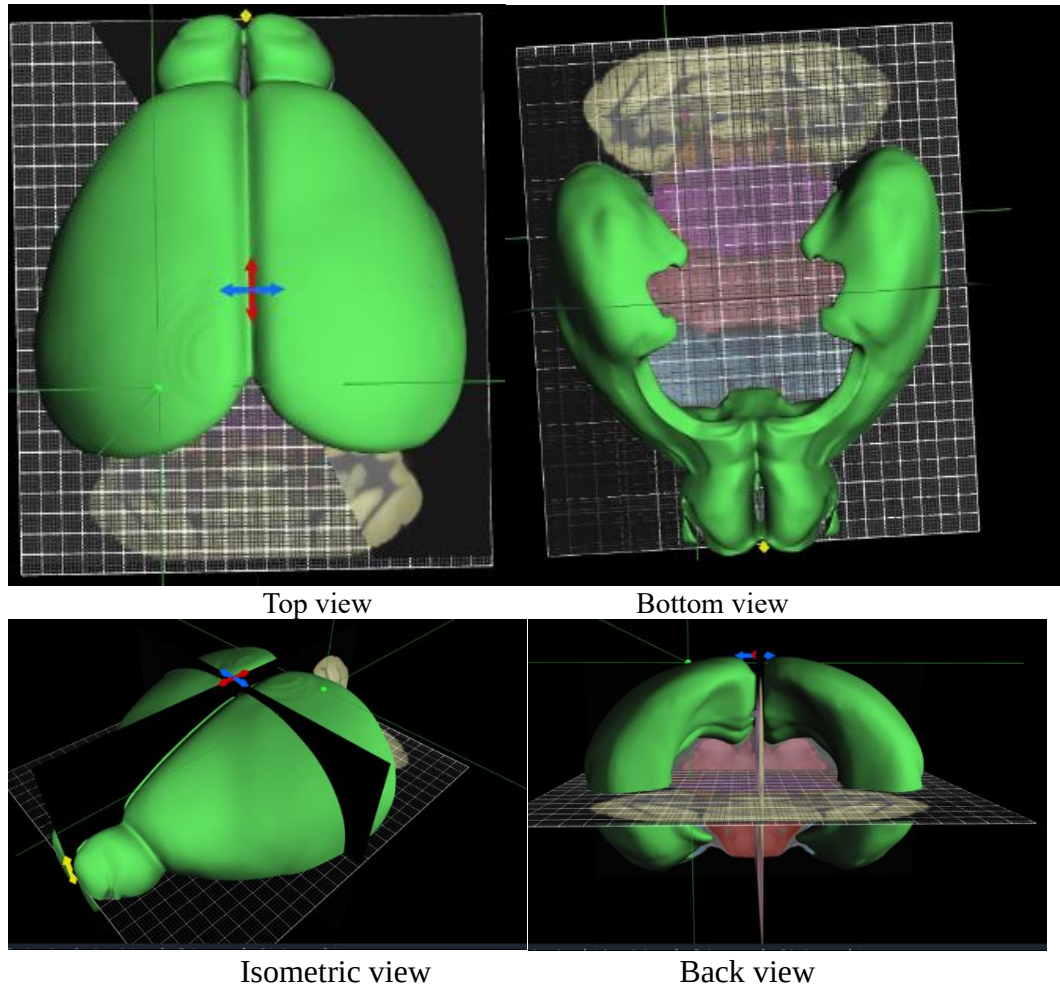


Figure 13: Cortex of mouse brain (Source: Allen rat brain atlas)

4.1.1. Types of feasible models utilized

4.1.1.1 Static setup:

In this setup, a rat is static and we apply an electric field through a parallel plate capacitor. If we keep distance 10 cm between plates of capacitor, then we need 1500 V potential difference between plates of capacitor which can be hazardous in terms of safety of personnel working in lab.



Figure 14: Expected look of prototype

4.1.1.2 Movable setup:

In this project, we aim to design a helmet-type setup that a rat can wear while freely moving in its cage. This setup is intended to achieve a high magnitude electric field within a voltage limit of 25V, thanks to the small size of the mouse head, making it a feasible option for in-vivo applications.

The project will progress through the following stages:

1. COMSOL Simulation for Electric Field Profile Analysis:

Model Development: Utilize the dimensions of the rat brain and the electrical properties of the skin, skull, cerebrospinal fluid (CSF), and brain tissues to develop an accurate 2D and 3D model.

2. Optimization of Voltage, Size, and Number of Electrodes:

Iterative Simulation:

- Conduct a series of simulations varying the width of electrodes and gaps between them.
- Optimize for uniform electric field distribution and minimal voltage requirement.

Graphical Analysis:

- Plot various parameters against width/gap ratios to determine the optimal design parameters. The following graphs were created to illustrate key aspects of the optimization process:
 - **Depth Where Last 150 V/cm is Occurred (Voltage applied = 10V):** Shows the depth from the electrode where the last occurrence of 150 V/cm electric field is observed.
 - **Maximum Electric Field on Cortex (Voltage applied = 10V):** Illustrates the maximum electric field achieved on the cortex surface for different width/gap ratios.
 - **Minimum Electric Field on Cortex Surface (Voltage applied = 10V):** Shows the minimum electric field on the cortex surface across various width/gap ratios.
 - **Uniformity:** Plots the electric field difference (indicating uniformity) against the width/gap ratios.
 - **Minimum Voltage Required to Get 150 V/cm on Cortex Surface:** Displays the minimum voltage needed to achieve a 150 V/cm electric field on the cortex surface for different width/gap ratios.
 - **Number of Electrodes:** Shows the number of electrodes required for each width/gap ratio.

This report covers the progress up to the COMSOL simulation and the analysis of the electric field profile and potential distribution across the electrodes. The methodology developed for the COMSOL simulation is detailed below:

• Methodology for COMSOL Simulation:

- Develop a detailed geometric model of the rat head, incorporating layers such as skin, skull, CSF, cortex, and white matter.
- Define the electrical properties of each layer based on experimental and literature values (in case of unavailability of mouse tissue properties, we used human tissue properties)
- Apply voltage to the electrodes in the model and simulate the resulting electric field distribution.
- Analyze the potential distribution and uniformity of the electric field across the cortex surface.
- Iterate the design by adjusting electrode width and gap ratios to achieve the optimal configuration for uniform electric field application.

Through this comprehensive approach, the project aims to develop a feasible and effective method for in-vivo applications in Alzheimer's disease research.

4.2. Model development for COMSOL simulation

To obtain uniform electric field of 150 V/cm concentrated on cortex, we need to simulate various combinations of electrodes and voltages applied on them. Different combinations were tried. Optimised among all of them is present in this report.

4.3. Software and version used

Simulation is carried out in COMSOL Multiphysics 5.5

4.4. Physics used and study

Add Electrostatics and select stationary study

4.5. Geometry

Parts of geometry:

1. Electrodes: To apply the voltage, we need electrodes.
2. Outmost sphere: Head including skin, skull bone, CSF, and brain (gray matter and white matter).

[For simplicity, cross-section of head is represented as circle]

3. Innermost sphere: White matter of brain

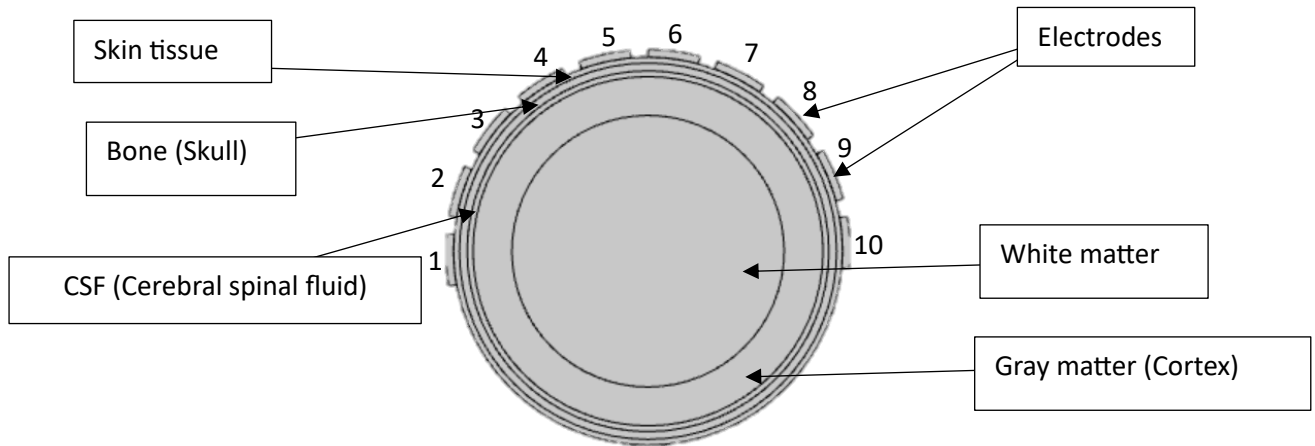


Figure 15: Parts of Geometry (Drawn in COMSOL)

Dimensions of geometry:

Electrodes:

- Total electrodes: 10
- Thickness of each electrode: 0.2 mm
- Sector angle: 15 degrees
- Spacing between electrodes: 5 degrees
- Total covered periphery of head: 200 degrees

Skin tissue around skull:

- Thickness: 0.15 mm

Skull (Bone tissue around brain):

- Thickness: 0.20 mm

Cerebral spinal fluid (CSF):

- Thickness: 0.15 mm

Gray matter (Cortex):

- Thickness: 1 mm

White matter (Middle part of brain):

- Radius: 3.5 mm

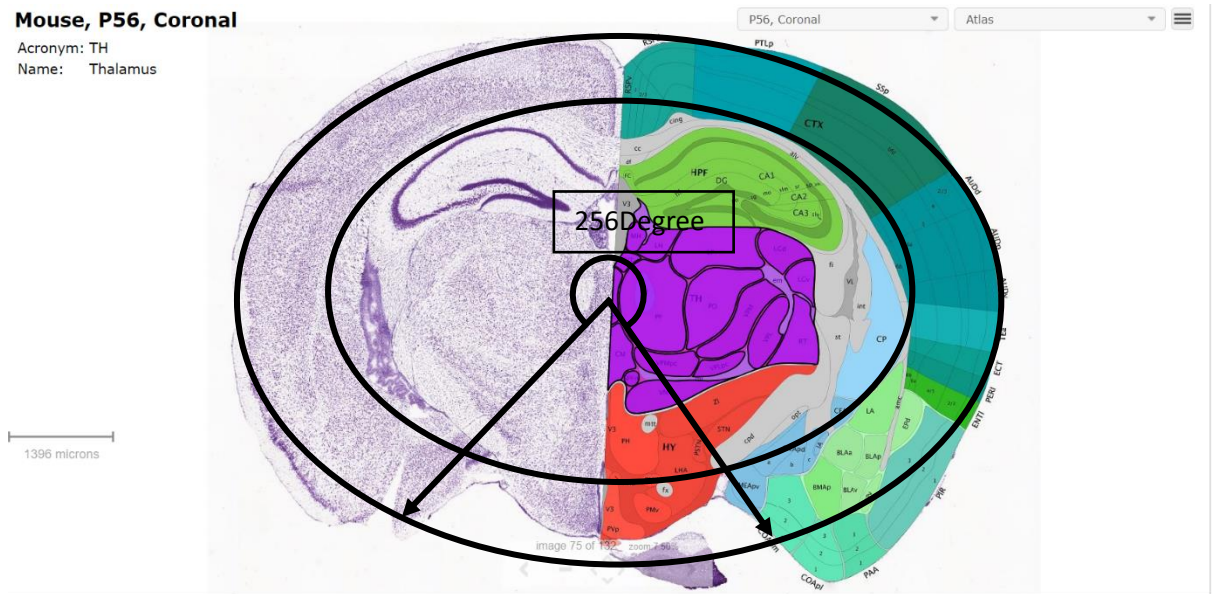


Figure 16: Radius for white matter and cortex is average of a and b axis of ellipses in picture shown
(Source: Allen brain institute)

4.6. Materials

Electrode material: Copper [Relative permittivity 1 & Electrical conductivity $5.998e7 \text{ S / m}$]

Availability of material: In-built material in COMSOL library.

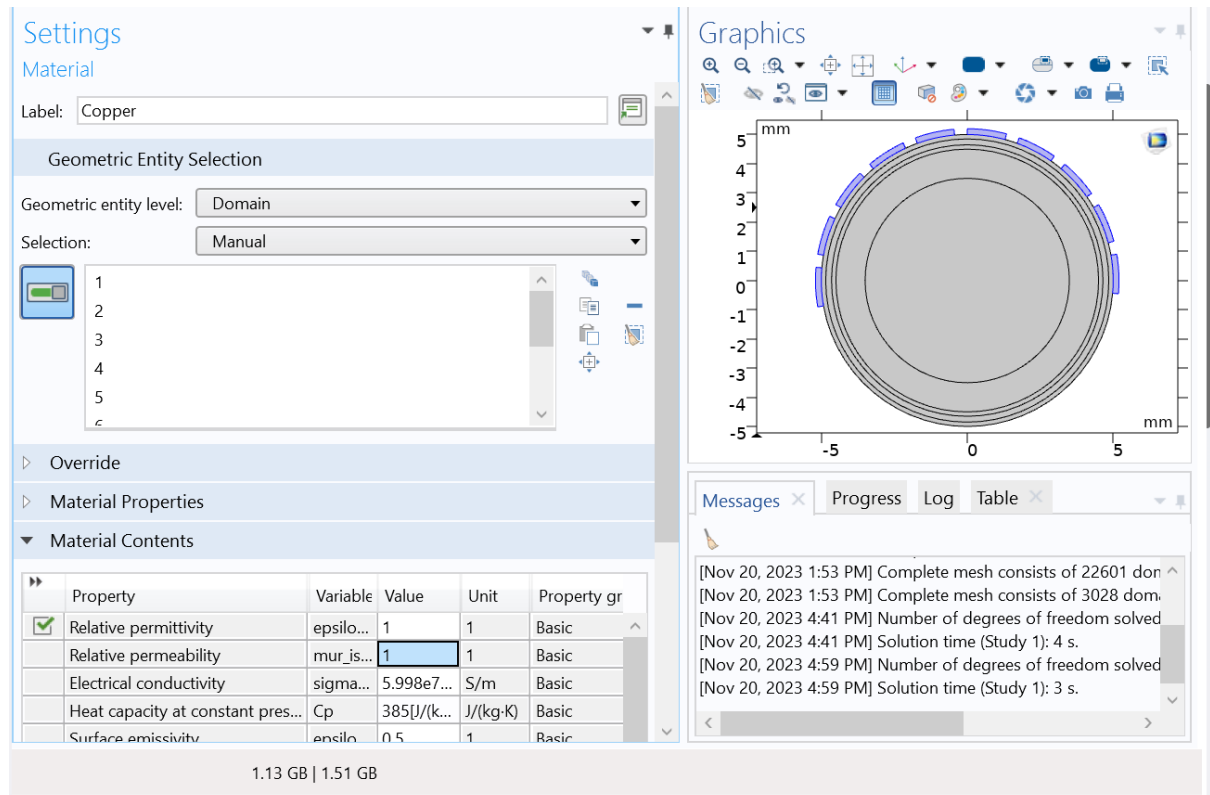


Figure 17: Properties of electrode material (Copper)

Skin tissue: Skin tissue [Relative permittivity is 20000 and Electrical conductivity is $.075 \text{ S} / \text{m}$ at 100 K Hz]

Availability of material: Created in COMSOL using properties described in (Gonzlez et al., 2009)

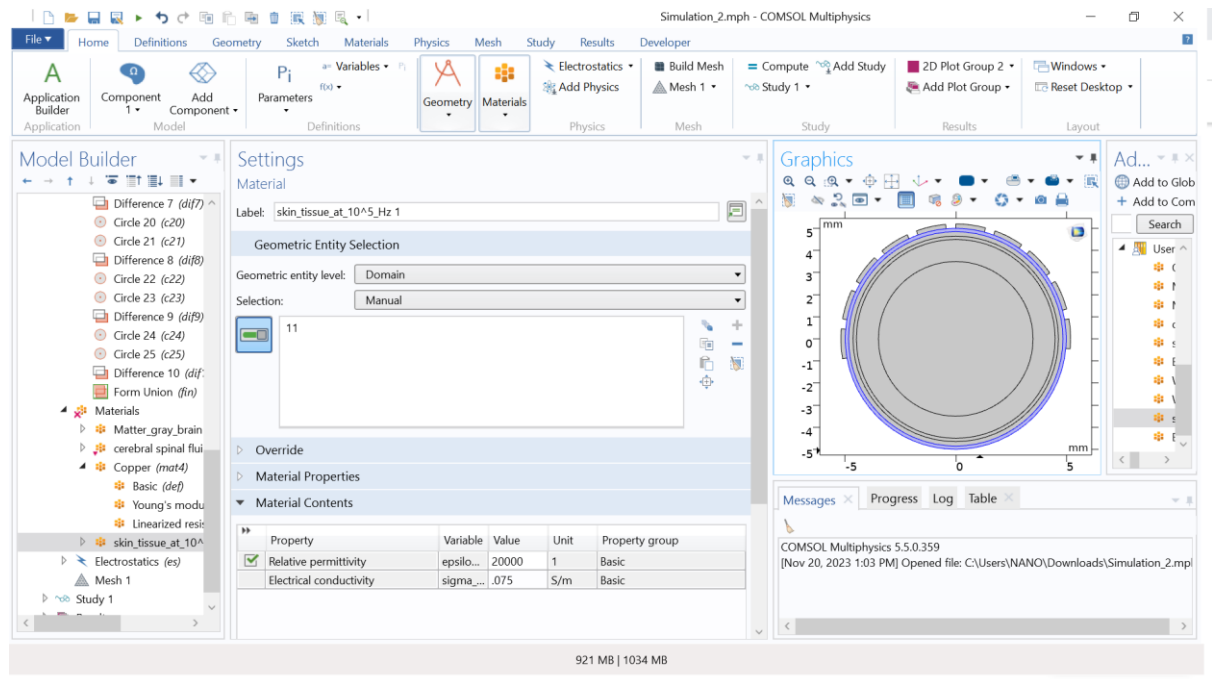


Figure 18: Properties of skin tissue material

Bone tissue: Bone tissue [Relative permittivity is 250 and Electrical conductivity is $.02 \text{ S} / \text{m}$ at 100 KHz]

Availability of material: Created in COMSOL using properties described in (Gonzlez et al., 2009)

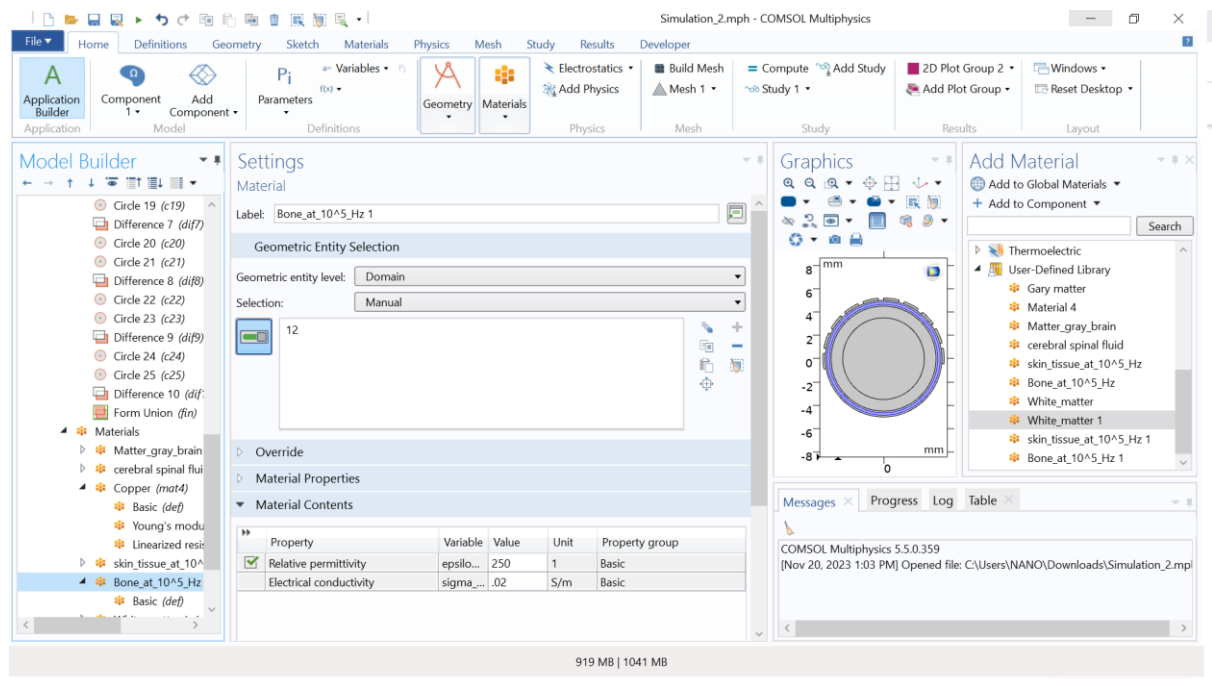


Figure 19: Properties of bone tissue material

CSF (Cerebral spinal fluid): Cerebral spinal fluid [Relative permittivity is 84.08 and Electrical conductivity is 2.143 S / m]

Availability of material: Created in COMSOL using properties described in (Gonzlez et al., 2009)

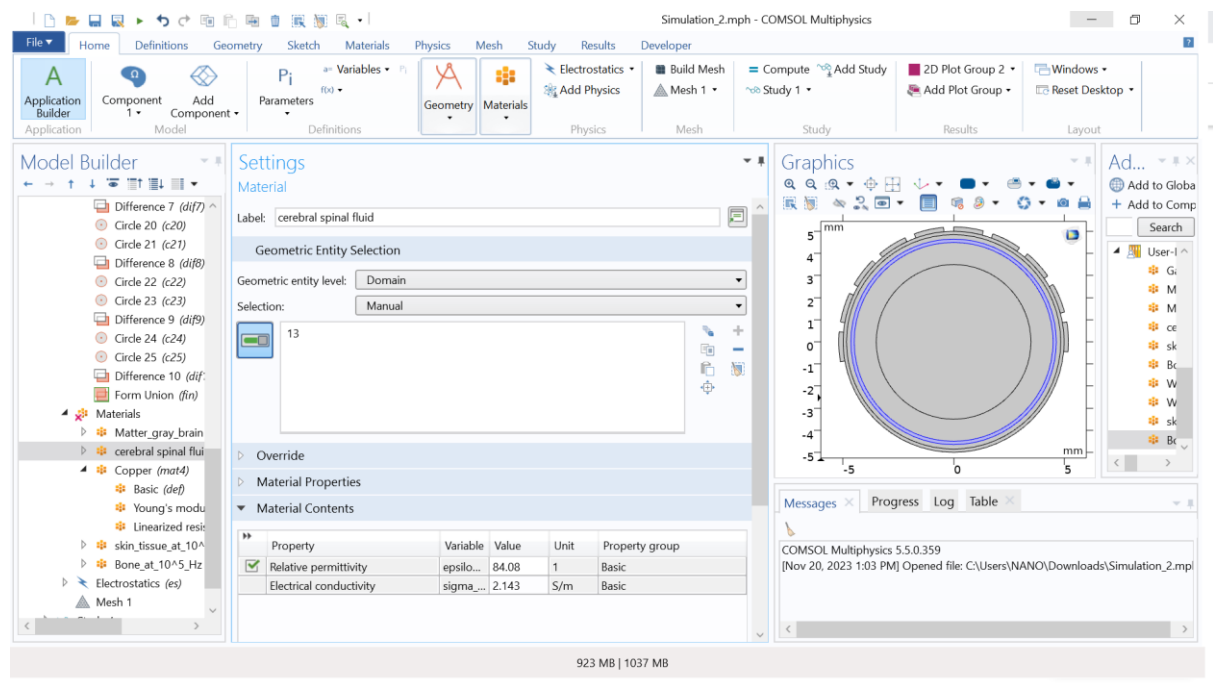


Figure 20: Properties of CSF material

Cortex (Gray matter): Gray matter [Relative permittivity is 76.61 and Electrical conductivity is 0.631 S / m]

Availability of material: Created in COMSOL using properties described in (Gonzlez et al., 2009)

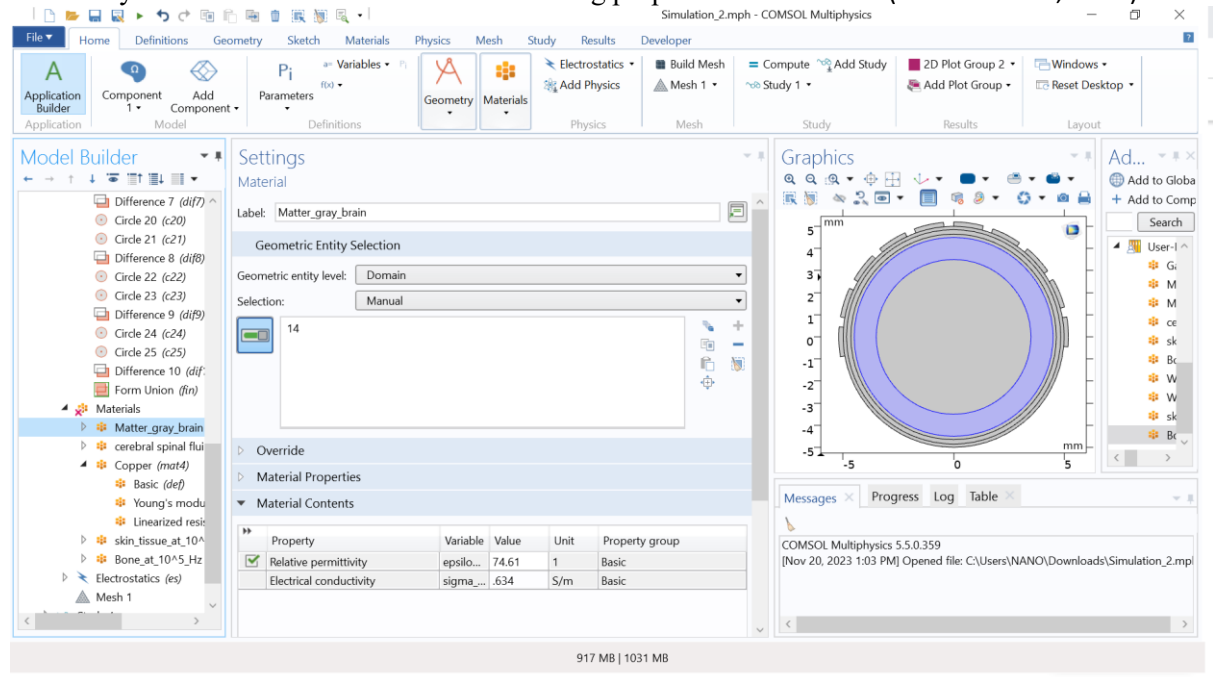


Figure 21: Properties of cortex tissue material

White matter: White matter [Relative permittivity is 52.57 and Electrical conductivity is $.342 \text{ S / m}$]

Availability of material: Created in COMSOL using properties described in (Gonzlez et al., 2009)

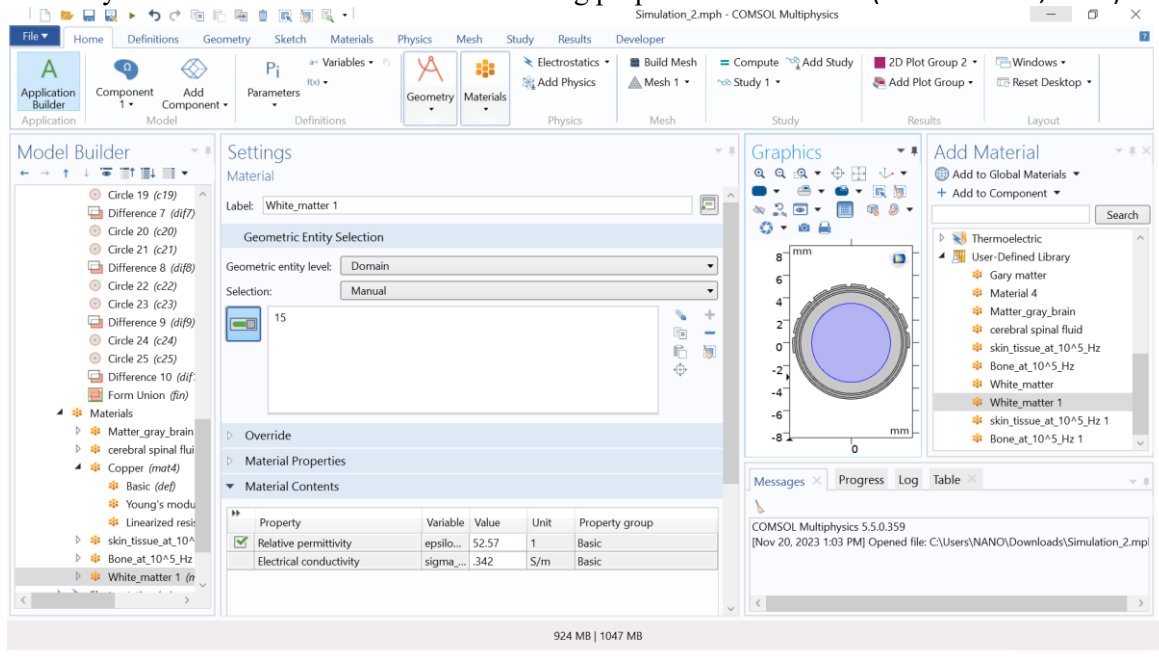


Figure 22: Properties of white matter

4.7. Electrostatics

1. Positive electrical potential:

Magnitude of potential: 10 V

Positive electrodes: 1, 3, 5, 7, 9 (Alternate manner)

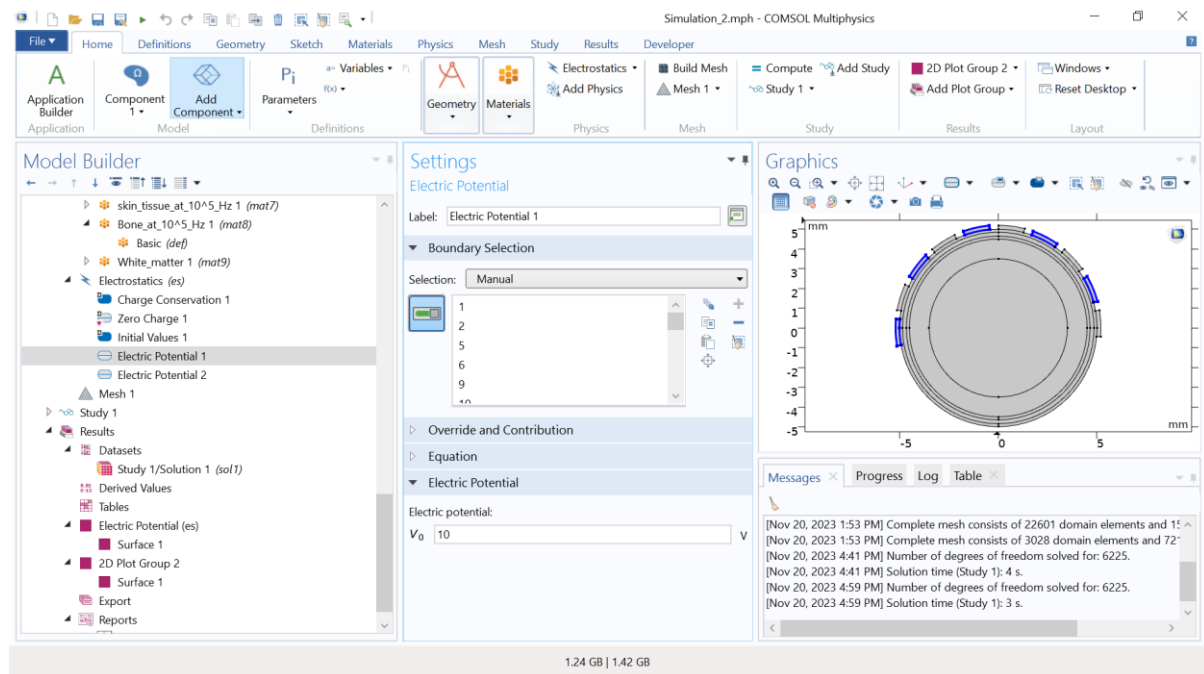


Figure 23: Positive electrodes

2. Negative electrical potential:

Magnitude of potential: -10 V

Positive electrodes: 2, 4, 6, 8, 10 (Alternate manner)

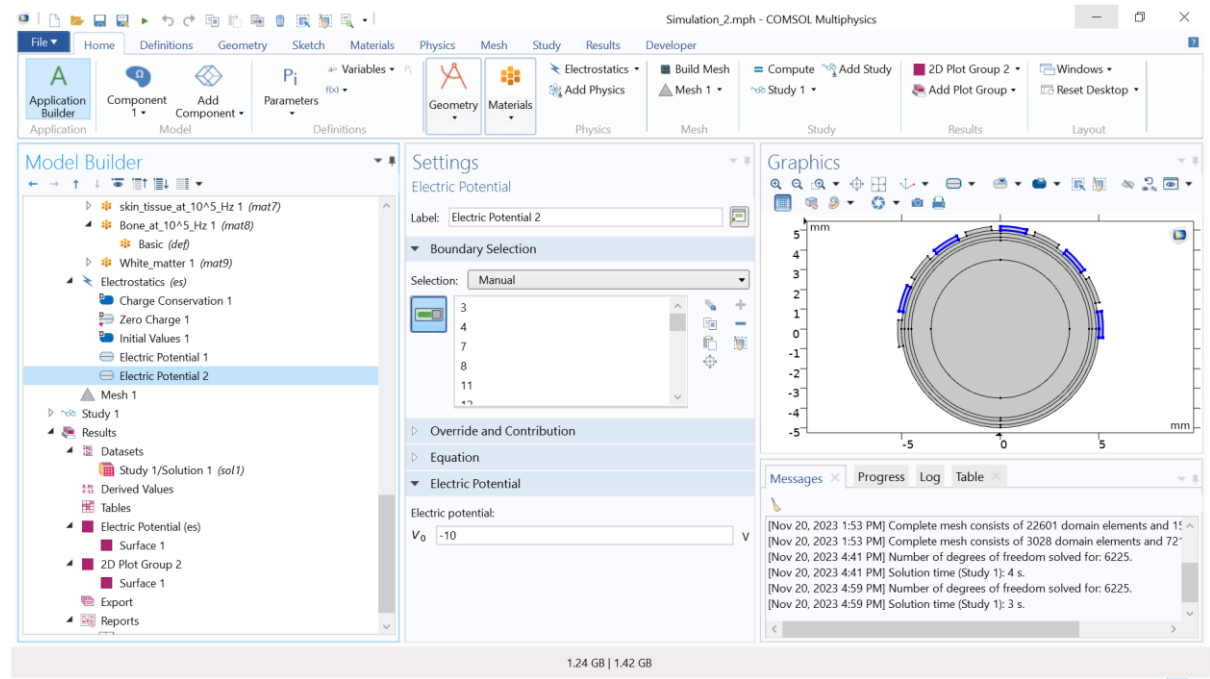


Figure 24: Negative electrodes

4.8. Meshing

Meshing is done separately for the geometry. COMSOL employs various automatic meshing algorithms that adapt to the geometry and physics of the problem to create suitable meshes. These algorithms aim to balance mesh quality, accuracy, and computational efficiency. Complete mesh consists of 3028 domain elements and 721 boundary elements.

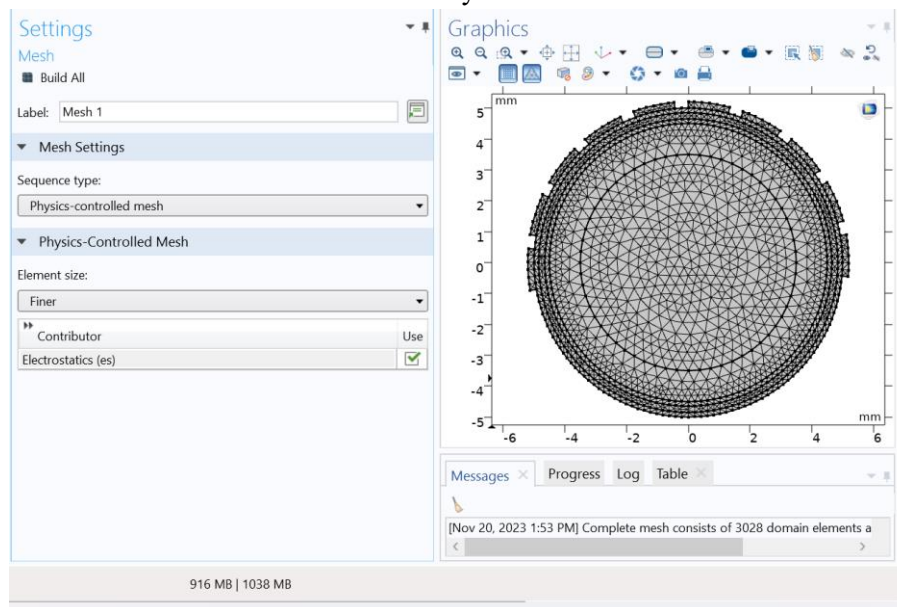


Figure 25: Meshing of geometry according to applied field

4.9. Study

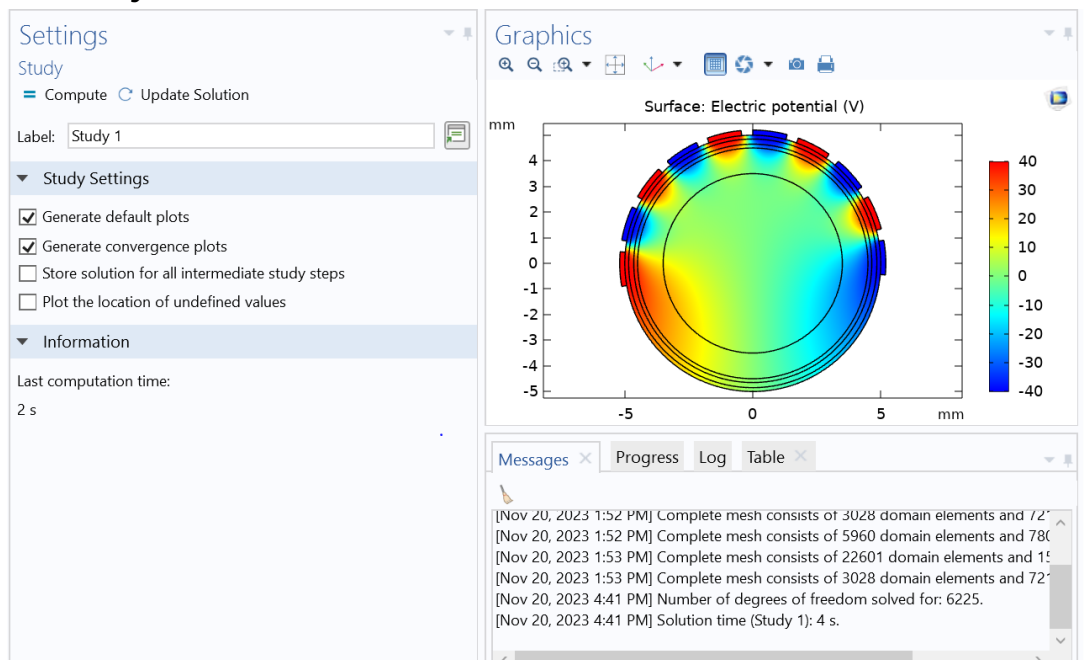


Figure 26: graphs according to applied physics

4.10. Electric field and electrical potential on basic model

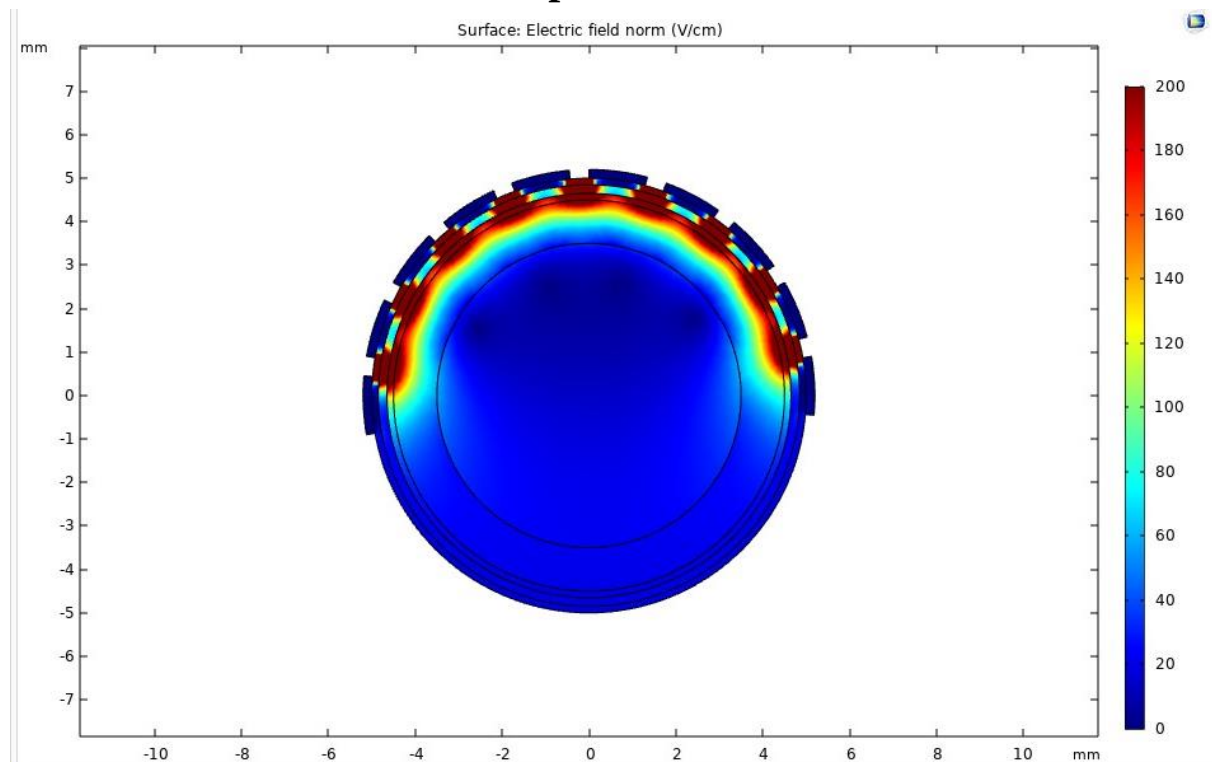


Figure 27: Electric Field distribution

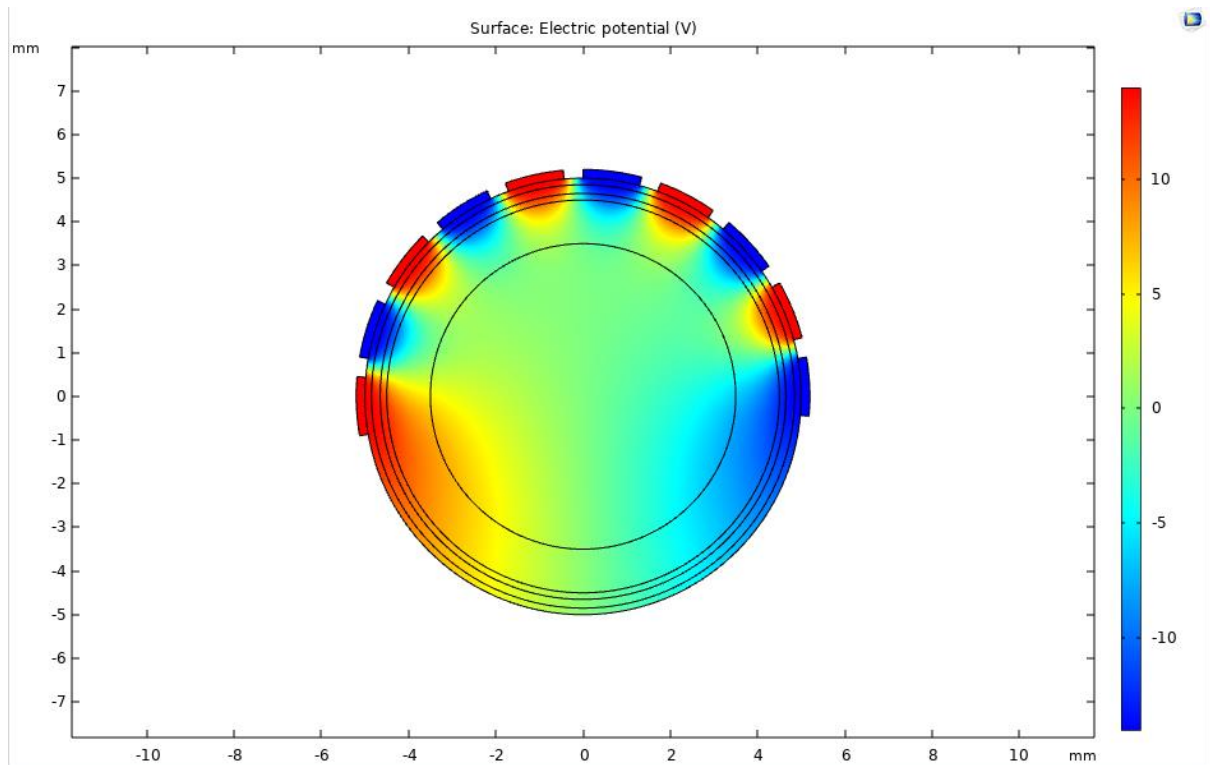


Figure 28: Electric potential distribution

4.11. Optimization of Electrode Design for Uniform Electric Field Delivery on basic model:

In the pursuit of optimizing the delivery of a uniform electric field to the cortex surface with minimal voltage, a systematic approach was employed to refine the basic 2D model of the mouse brain. This optimization involved varying the width of the electrodes and the gaps between them. The goal was to achieve a uniform electric field distribution across the target area while minimizing the required input voltage.

1. Initial Model Setup:

- The initial 2D model, created in COMSOL Multiphysics, included the basic anatomical features of the mouse brain.

2. Electrode Design Parameters:

- A set of initial dimensions for electrode width and gap was chosen, denoted as width/gap ratio.

3. Simulation and Iteration:

- The model was simulated with the initial electrode design.
- Parameters such as electric field distribution and voltage requirements were noted.
- The width and gap ratio were then systematically varied in subsequent simulations.
- Each iteration involved:
 - Depth (Where last 150 V/cm is there) at 10 V.
 - Maximum Electric field on cortex when 10 V applied.
 - Minimum electric field on cortex surface when 10 V applied.
 - Uniformity (Difference between minimum and maximum).
 - Minimum voltage required to get 150 V/cm on cortex surface.
 - Number of electrodes.

4. Data Collection and Analysis:

- For each iteration, the parameters were meticulously noted.

- The results were plotted on a graph to visualize the relationship between electrode width, gap, and electric field uniformity.
- Analysis of the plots helped identify the optimal electrode dimensions that produced the desired uniform electric field with the minimum voltage.

5. Final Optimized Design:

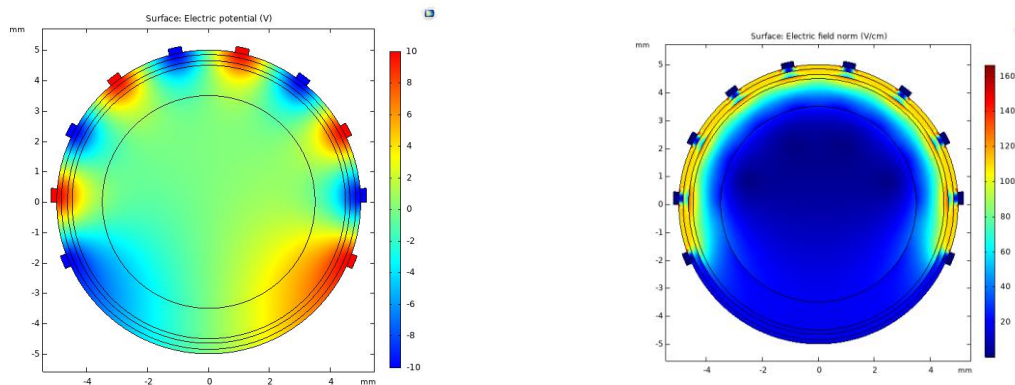
- The optimal width and gap dimensions were determined based on the iterative simulations and analysis.
- These optimized dimensions were then incorporated into the final model.

4.12 Systematic Representation

To visually represent the optimization process, the following pictures are shown:

Model 1: Parameters:

1. Width of electrode: 5 Degree
2. Width of gap: 20 Degree
3. Width of electrode/Width of gap: .25
4. Number of electrodes: 10
5. Angle covered: 230 Degree



Model 2: Parameters: *Figure 29: Electric potential distribution and Electric field distribution for Model 1*

1. Width of electrode: 5 Degree
2. Width of gap: 10 Degree
3. Width of electrode/Width of gap: .5
4. Number of electrodes: 17
5. Angle covered: 245 Degree

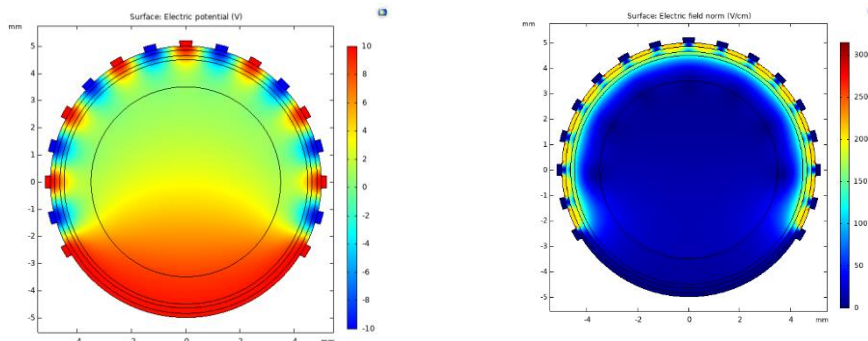


Figure 30: Electric potential distribution and Electric field distribution for Model 2

Model 3: Parameters:

1. Width of electrode: 5 Degree
2. Width of gap: 6.666 Degree
3. Width of electrode/Width of gap: .75
4. Number of electrodes: 21
5. Angle covered: 238.33 Degrees

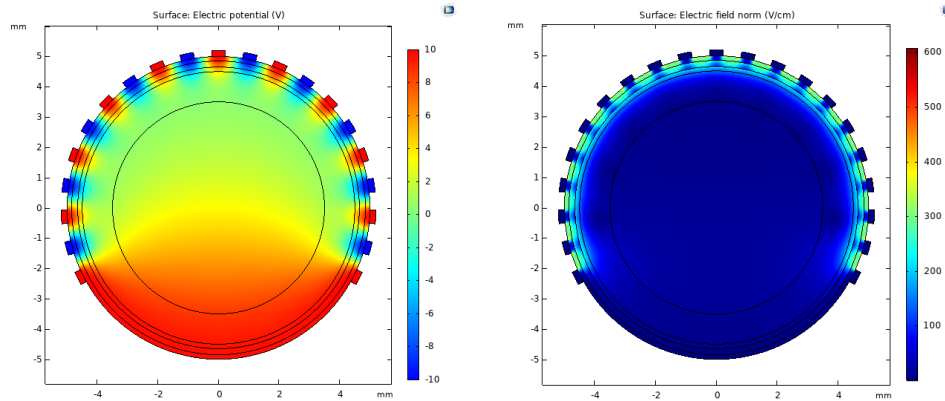


Figure 31 Electric potential distribution and Electric field distribution for Model 3

1. Width of electrode: 5 Degree
2. Width of gap: 5 Degree
3. Width of electrode/Width of gap: 1
4. Number of electrodes: 25
5. Angle covered: 245 Degree

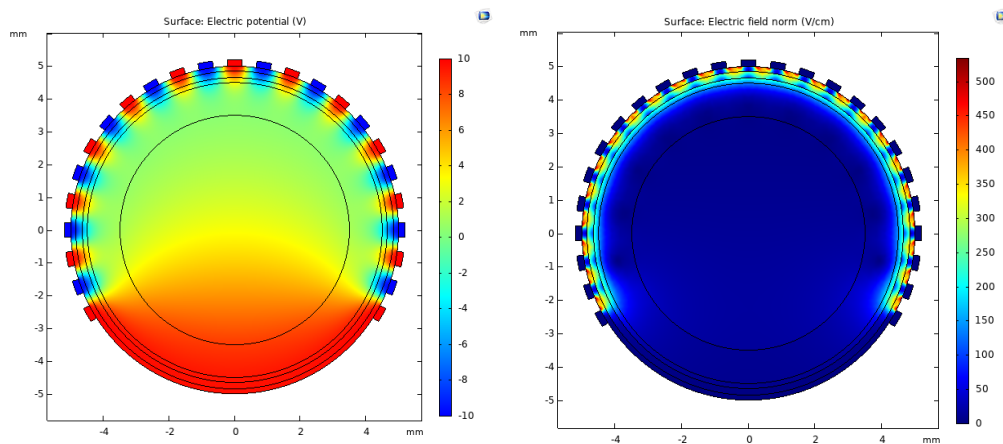


Figure 32: Electric potential distribution and Electric field distribution for Model 4

Model 5: Parameters:

1. Width of electrode: 5 Degree
2. Width of gap: 4 Degree
3. Width of electrode/Width of gap: 1.25
4. Number of electrodes: 28
5. Angle covered: 248 Degree

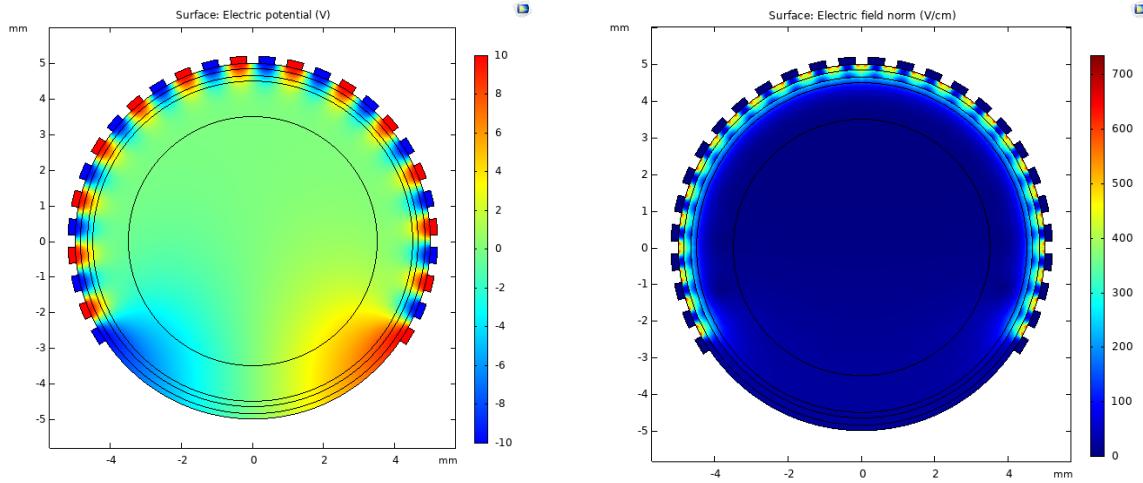


Figure 33: Electric potential distribution and Electric field distribution for Model 5

Model 6: Parameters:

1. Width of electrode: 5 Degree
2. Width of gap: 3.33 Degree
3. Width of electrode/Width of gap: 1.50
4. Number of electrodes: 30
5. Angle covered: 246.66

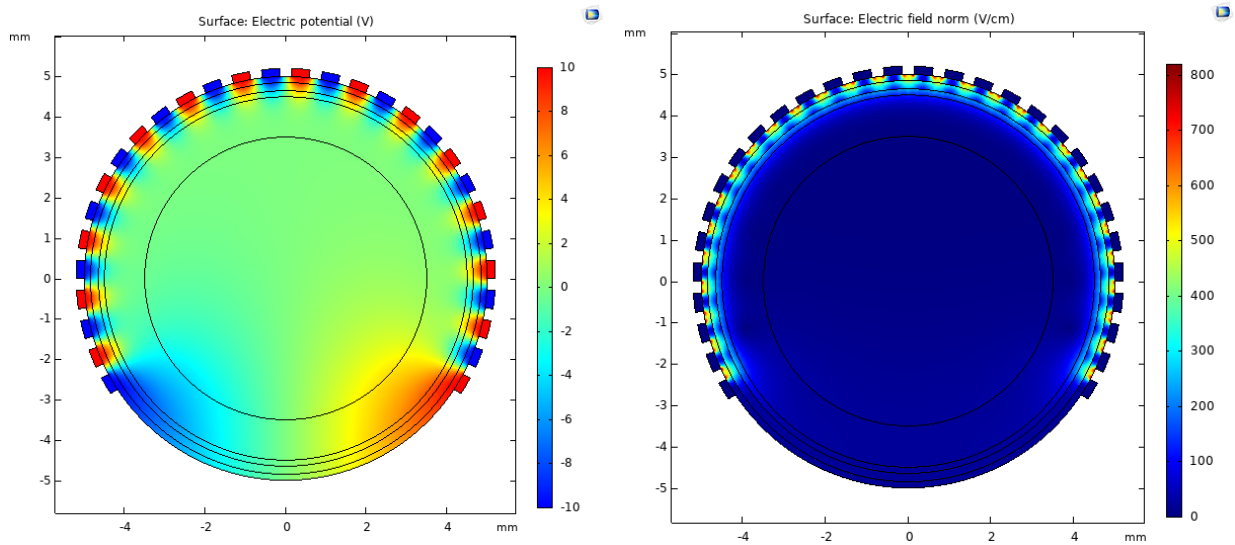


Figure 34: Electric potential distribution and Electric field distribution for Model 6

Model 7: Parameters:

1. Width of electrode: 5 Degree
2. Width of gap: 2.85
3. Width of electrode/Width of gap: 1.75
4. Number of electrodes: 32
5. Angle covered: 248.57 Degree

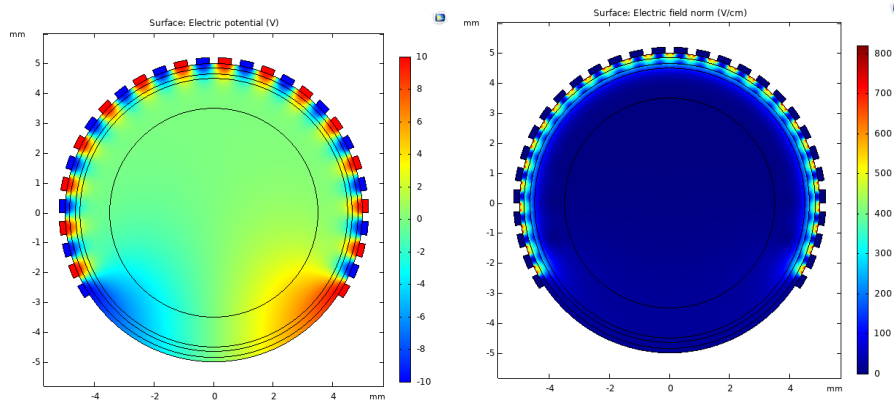


Figure 35: Electric potential distribution and Electric field distribution for Model 7

4.13 Graphical Analysis of Optimization Parameters:

The optimization process involved varying the width of the electrodes and the gaps between them to achieve a uniform electric field with minimal voltage requirements. The following graphs illustrate the results of this optimization process:

1. **Depth Where Last 150 V/cm is Occurred (Voltage applied = 10V):** This graph shows the depth from the electrode where the last occurrence of a 150 V/cm electric field is observed.

Analysis:

- As the width/gap ratio increases, the depth initially increases, reaching a peak around a ratio of 1, and then decreases.
- The maximum depth is observed at a width/gap ratio of approximately 1, indicating the optimal ratio for achieving the desired field depth.

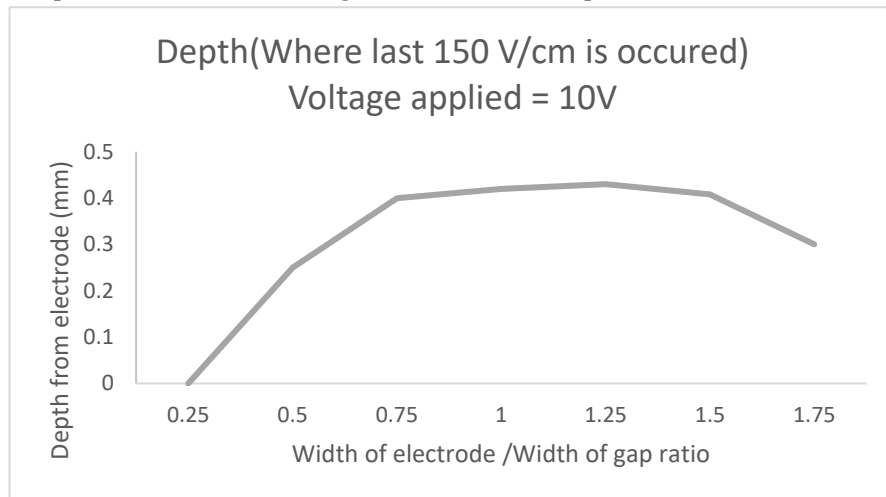


Figure 36: Electric field penetration

2. **Maximum Electric Field on Cortex (Voltage applied = 10V):** This graph illustrates the maximum electric field achieved on the cortex surface for different width/gap ratios.

Analysis:

- The electric field peaks around a width/gap ratio of 1, indicating that this ratio provides the highest field strength.
- Beyond this ratio, the field strength decreases, suggesting that higher or lower ratios are less effective.

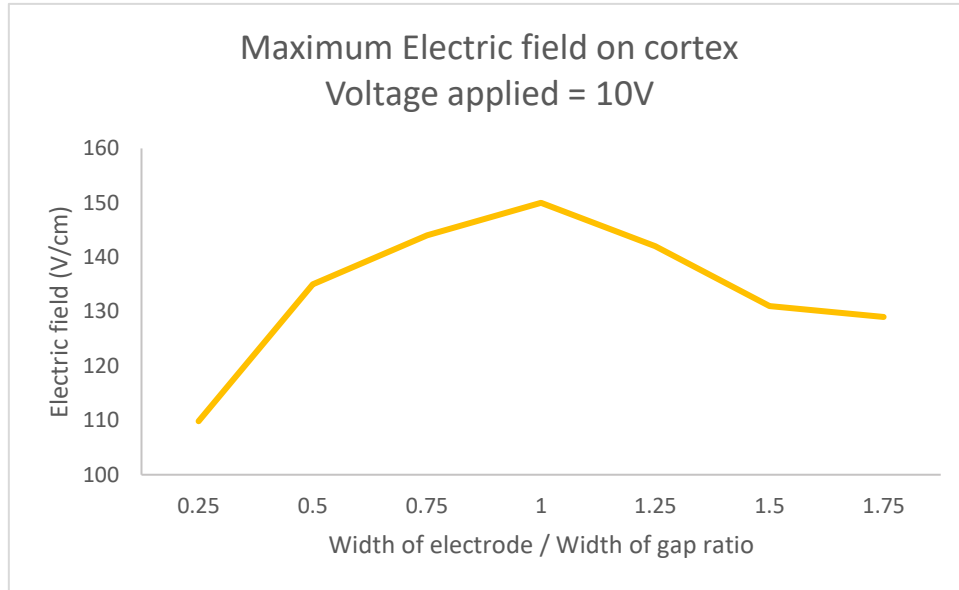


Figure 37: Maximum Electric field on cortex surface

3. **Minimum Electric Field on Cortex Surface (Voltage applied = 10V):** This graph shows the minimum electric field on the cortex surface across various width/gap ratios.

Analysis:

- The minimum electric field increases up to a width/gap ratio of approximately 0.75 and then stabilizes.
- This indicates that a ratio around 0.75 to 1 provides a stable and sufficient minimum field strength.

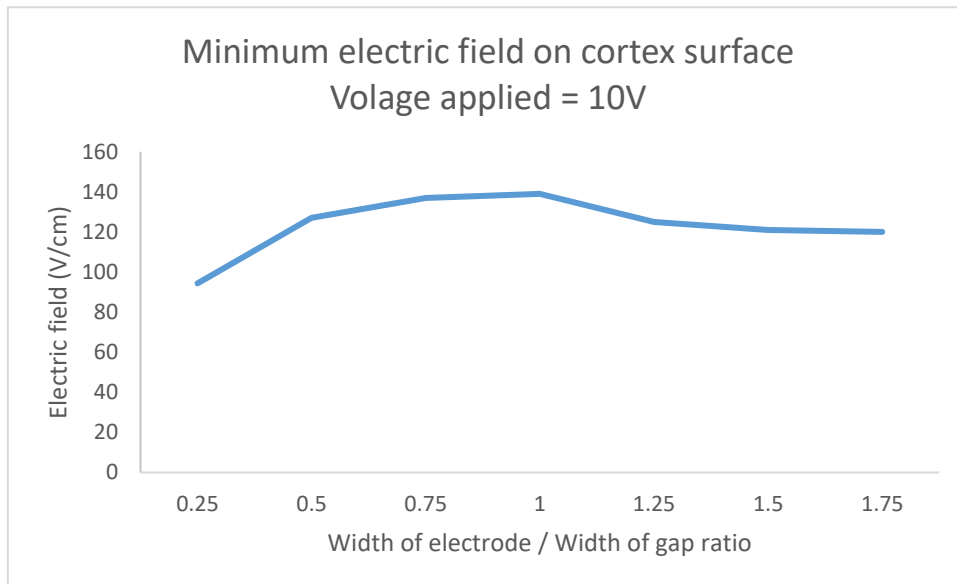


Figure 38: Minimum Electric field on cortex surface

4. **Uniformity:** This graph plots the electric field difference (indicating uniformity) against the width/gap ratios.

Analysis:

- As the difference between the maximum and minimum electric fields increases, the uniformity decreases.

- The optimal uniformity is achieved around width/gap ratios of 0.5 and 1.25, where the electric field difference is minimized, indicating more consistent field strength.

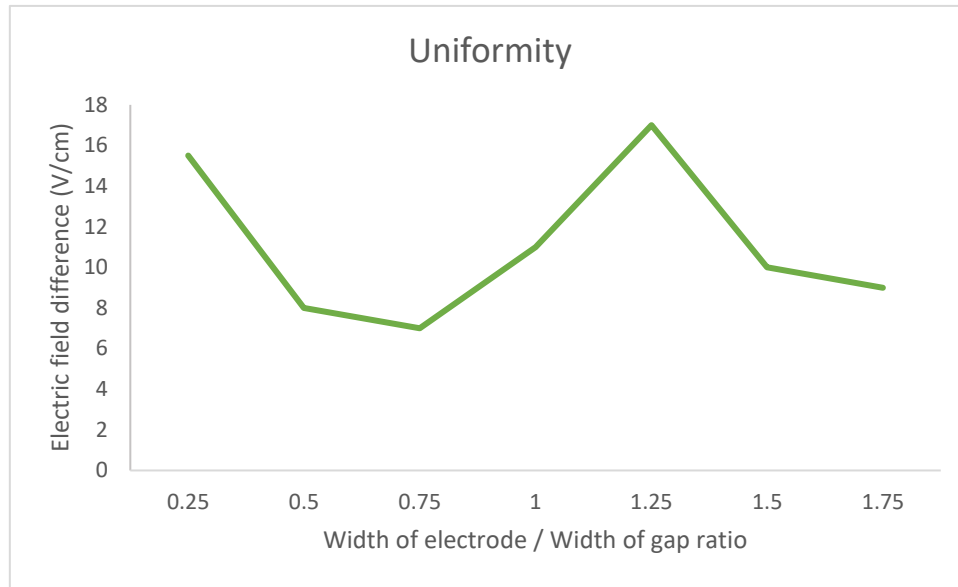


Figure 39: Uniformity of Electric field distribution on cortex surface

- 5. Minimum Voltage Required to Get 150 V/cm on Cortex Surface:** This graph displays the minimum voltage needed to achieve a 150 V/cm electric field on the cortex surface for different width/gap ratios.

Analysis:

- The minimum voltage requirement decreases up to a width/gap ratio of 1, then starts to increase again.
- The lowest voltage requirement is observed around a width/gap ratio of 1, making it the most efficient ratio for voltage application.

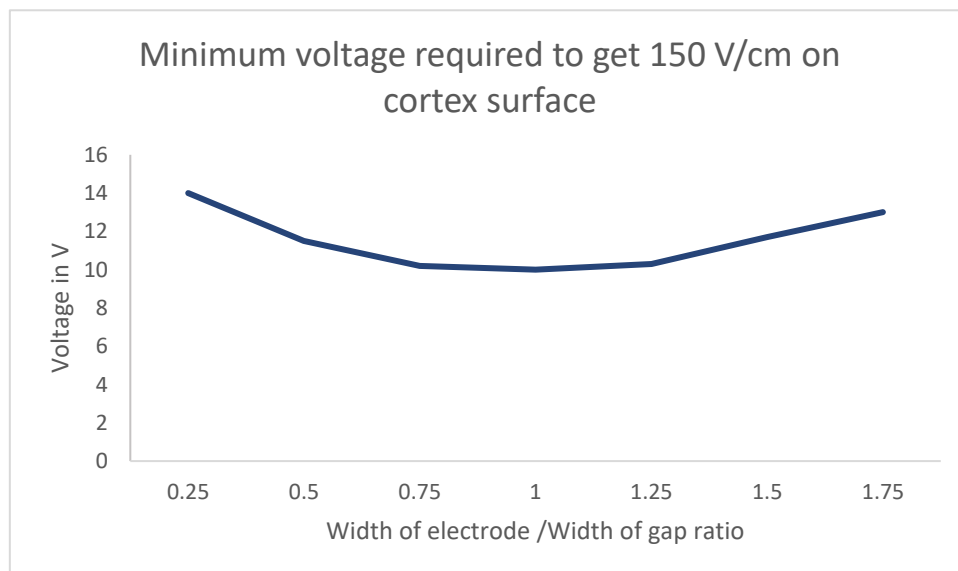


Figure 40: Minimum voltage required to get 150 V/cm on cortex surface

- 6. Number of Electrodes:** This graph shows the number of electrodes required for each width/gap ratio.

Analysis:

- The number of electrodes increases with the width/gap ratio.
- The most significant increase occurs between ratios 0.75 to 1.25, indicating a trade-off between the number of electrodes and the optimal electric field.

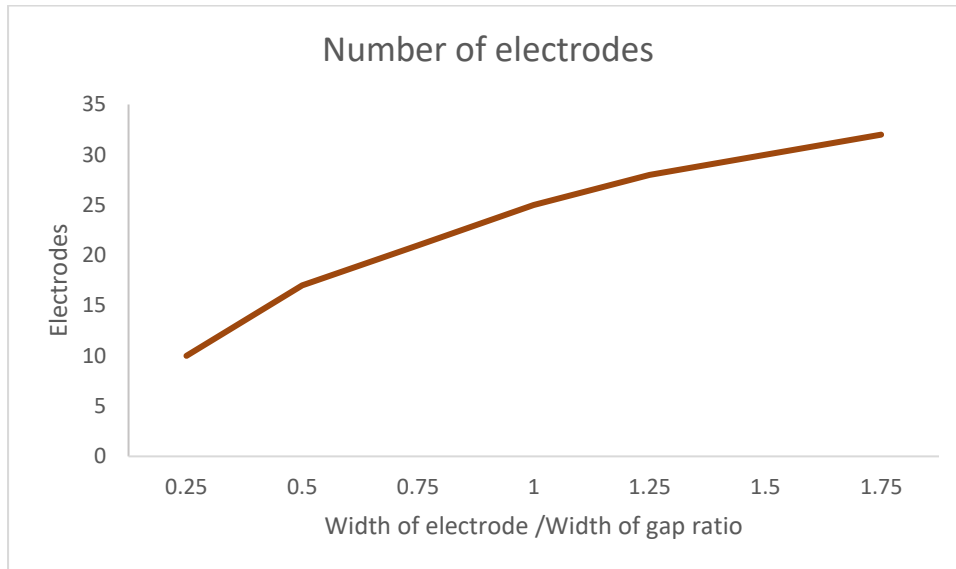


Figure 41: Number of electrodes

The optimal width/gap ratio for achieving a uniform electric field with minimal voltage requirement and efficient electrode usage is around 1. This ratio provides the best balance between depth, maximum and minimum electric fields, uniformity, and voltage requirements. These findings guide the design of the helmet-type setup for in-vivo applications, ensuring effective and feasible implementation of electric field application for Alzheimer's disease research.

Final Optimized Electrode Design:

The electric field norm and electric potential images for the width of electrodes to width of gap ratio of 1 present a comprehensive and highly effective model for in-vivo applications. This ratio provides a harmonious balance between high electric field intensity and uniform potential distribution, crucial for precise and controlled stimulation of the target area. The electric field norm image clearly illustrates the intense field concentration at the edges of the electrodes, creating a consistent ring of high intensity around the perimeter. This intense field gradually diminishes inward, ensuring a well-distributed field across the desired area, which is critical for achieving effective stimulation. Notably, within the innermost circle, the electric field is almost zero, which helps in safeguarding the central region from excessive stimulation, thereby protecting delicate tissues.

The electric potential distribution further underscores the efficacy of this model. The smooth gradient from high potential at the edges to lower potential towards the center indicates a successful voltage application, maintaining symmetry and uniformity around the circular configuration. This controlled potential gradient is essential for creating a stable and predictable electric field, which is necessary for reliable in-vivo experiments. The regions of highest potential, transitioning seamlessly to lower potential areas, confirm the electrode design's capability to generate a desired potential distribution efficiently.

In summary, the width of electrodes to width of gap ratio of 1 emerges as the optimal configuration for creating a uniform electric field and potential distribution. This balance ensures that the electric field penetrates effectively into the target area, which is vital for applications like stimulating the rat brain. Moreover, the model demonstrates voltage efficiency, achieving the required electric field strength with minimal applied voltage. This efficiency not only enhances the effectiveness of the setup but also ensures the safety and feasibility of the design for in-vivo research on neurological conditions such as Alzheimer's disease. The images collectively confirm that this model is not only theoretically sound but also practically viable for achieving precise and controlled electric field applications in experimental settings.

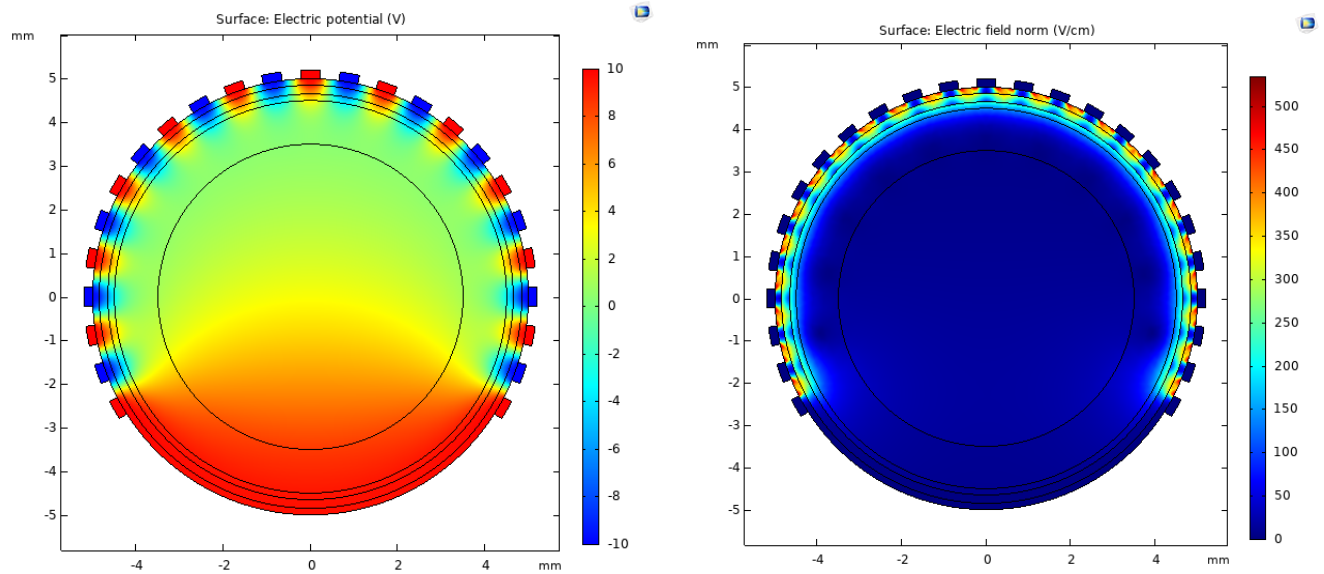


Figure 42: Electric potential distribution and Electric field distribution for optimized model

Chapter 5

5. Advanced model

5.1. 3D Printing of Mouse Brain and Skull Models

To enhance the understanding of the spatial dimensions and structural relationships between the mouse brain and skull, 3D models were created and printed using a Formlabs SLA 3D printer, which utilizes stereolithography technology. The .stl file for the mouse brain model was sourced from the Allen Brain Institute, while the .stl file for the mouse skull was obtained from alternative sources. The high-resolution 3D prints provided a tangible representation, facilitating a more comprehensive grasp of the anatomical features and their spatial orientation. This hands-on approach proved invaluable in appreciating the intricate details and scale of the mouse brain in relation to its skull.

5.1.1. Overview of 3D Printed Models:

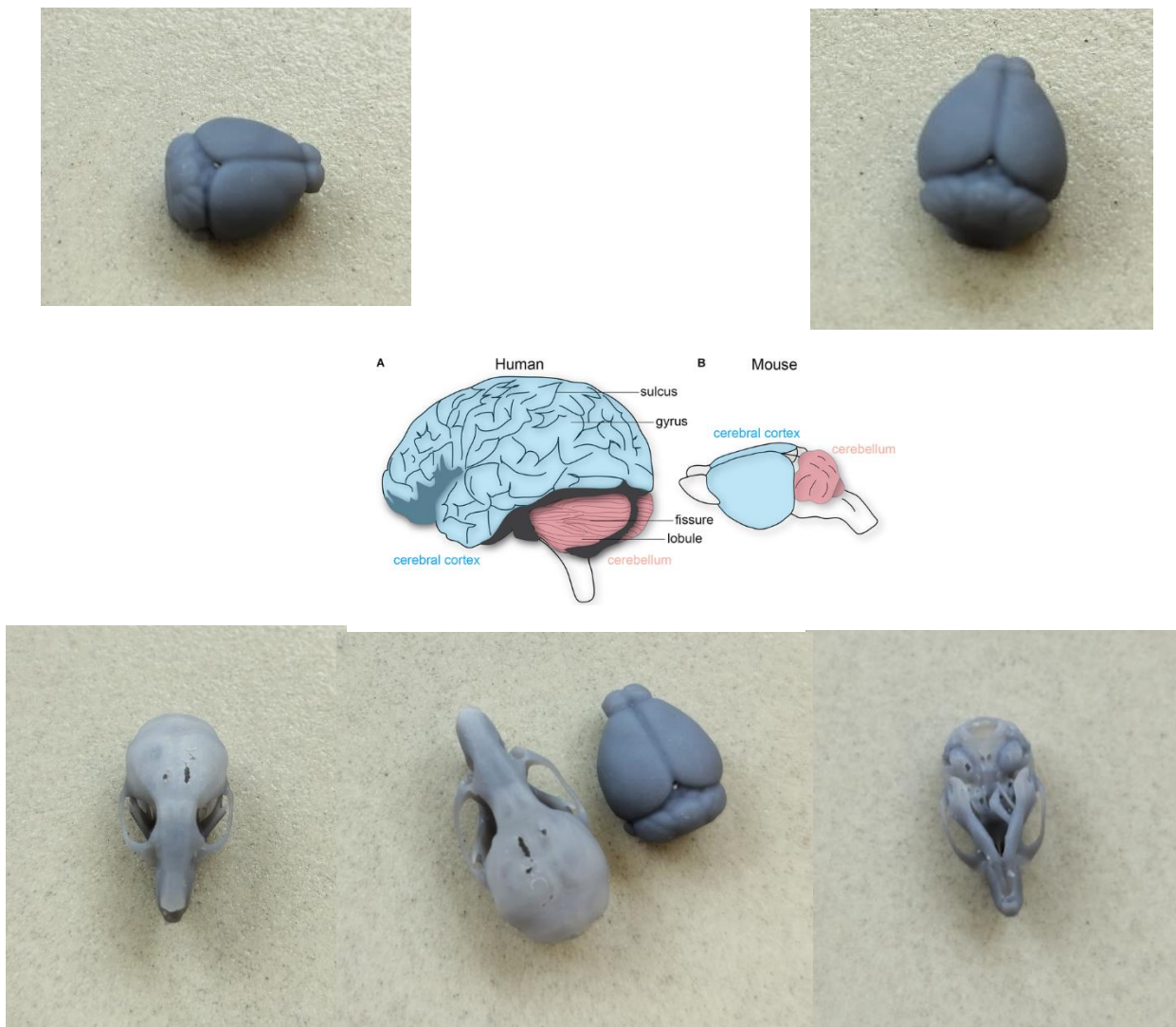


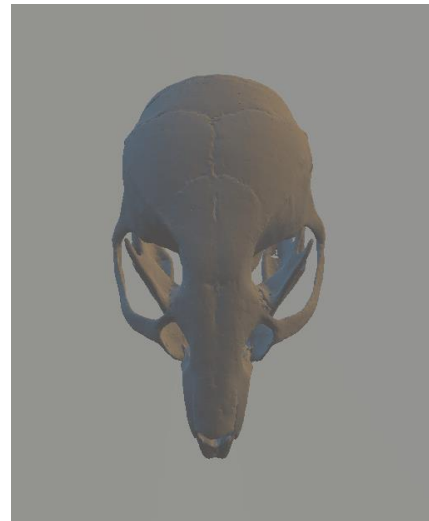
Figure 43: 3D printed mouse skull and brain



(a): Comparison of dimensions with scale



(b): View of 3D model of mouse brain in computer



(c): View of 3D model of mouse skull in computer

Figure 44: Dimension comparison and 3D model views in computer

5.1.2. Printing Process:



(a) Formlabs Stereolithography-based printer



(b) Pre-processed 3D prints of mouse skull and brain

Figure 45: 3D printer and preprocessed 3D models

5.2. Advanced model:

In the initial phase of this project, a basic 2D model of the mouse brain was created using COMSOL Multiphysics to capture its essential anatomical features and dimensions. This preliminary model served as a foundational representation for understanding the brain's basic structure. Subsequently, the model's complexity was enhanced by incorporating additional elements, such as the scalp layer, and by taking into account the maximum dimensions. These improvements provided a more comprehensive and realistic representation of the mouse brain, allowing for a detailed examination of its spatial relationships and anatomical intricacies.

In this advanced phase, the focus shifted towards simulating a device capable of delivering an electric field of 150 V/cm to the cortex surface. To achieve this, various electrode patterns were designed and integrated into the model. Additionally, to fabricate the flexible printed circuit board (PCB) necessary for this device, a polyimide layer was added. This inclusion of the polyimide layer further enhanced the model, ensuring the design was suitable for real-world fabrication and application. This progression was crucial for developing a more accurate and functional model to aid in the precise delivery of electric fields for experimental and therapeutic purposes.

To optimize the device's performance, the basic model was further refined by varying the width of the electrodes and the gaps between them. This optimization aimed to achieve a uniform electric field distribution across the cortex surface while minimizing the required voltage. By systematically adjusting these parameters, the model was able to identify the optimal configuration that delivered a consistent electric field with the least amount of voltage, enhancing the device's efficiency and effectiveness. This iterative optimization process was crucial for developing a practical and reliable device for precise electric field delivery in experimental and therapeutic applications.

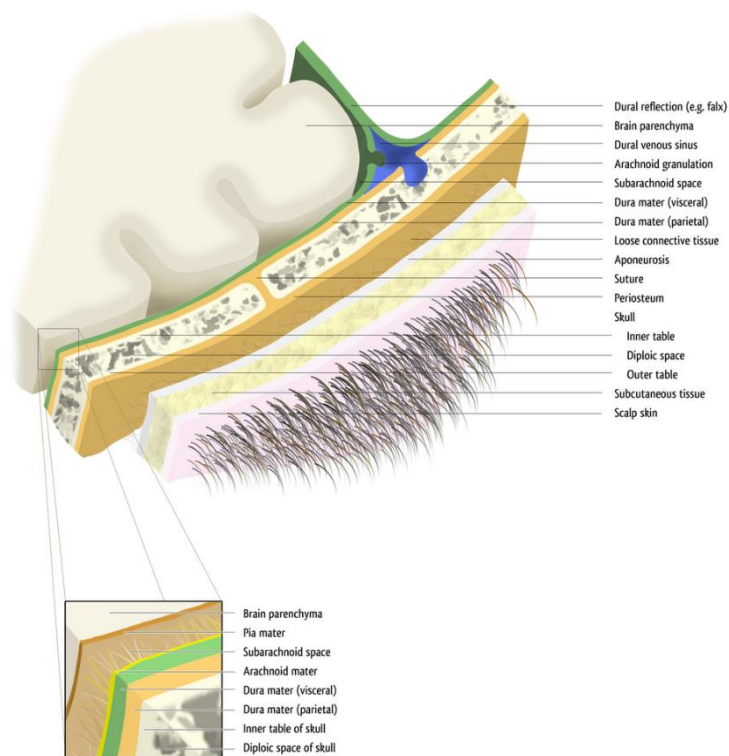


Figure 46: Detailed representation of layers present in head

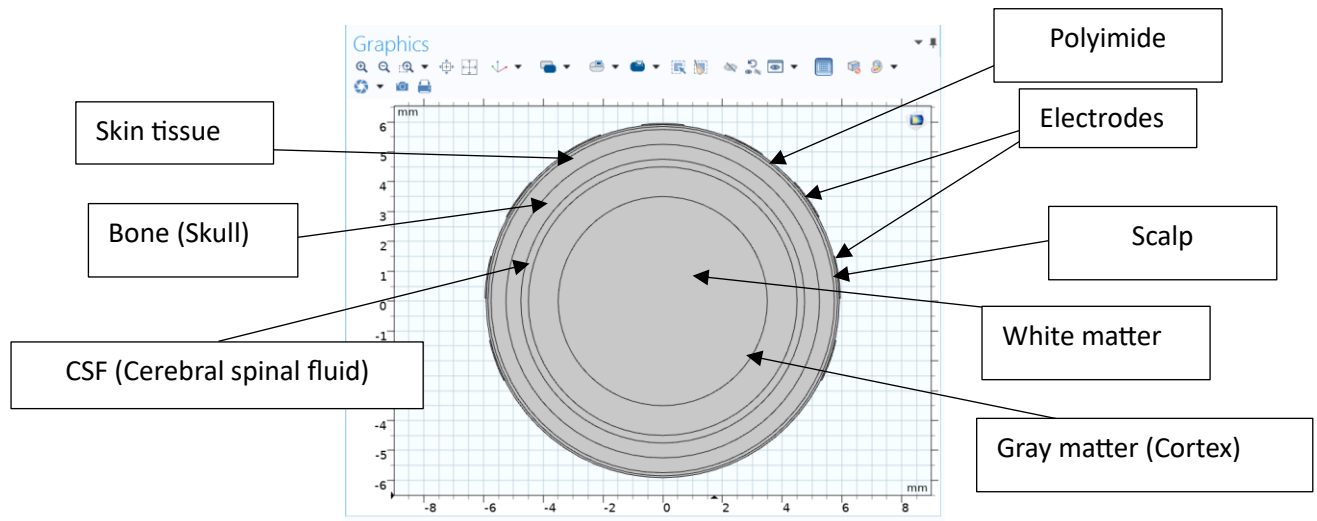


Figure 47: Geometry of advanced model

Angle to be covered: 256 Degree

Thickness of polyimide sheet:	0.065 mm
Thickness of fur:	0.1 mm
Thickness of skin:	0.5 mm
Thickness of skull:	0.5 mm
Thickness of CSF:	0.25 mm
Total thickness to penetrate EF into the cortex:	1.415 mm

	Copper electrode thickness:	1 oz or	.0348 mm
	Minimum trace width	.01 inch or	.254 mm
	So, minimum width of electrodes in degrees		0.0007494735926

Material properties:

Serial no.	Material name	Conductivity(S/m)	Relative permittivity
1.0	Polyimide sheet	0.0000000001	3.4
2.0	Fur	0.49	72.9
3.0	Skin	1.46	38
4.0	Skull	0.39	11.38
5.0	CSF	2	109
6.0	Gray matter	1.03	55.5
7.0	White matter	0.47	37

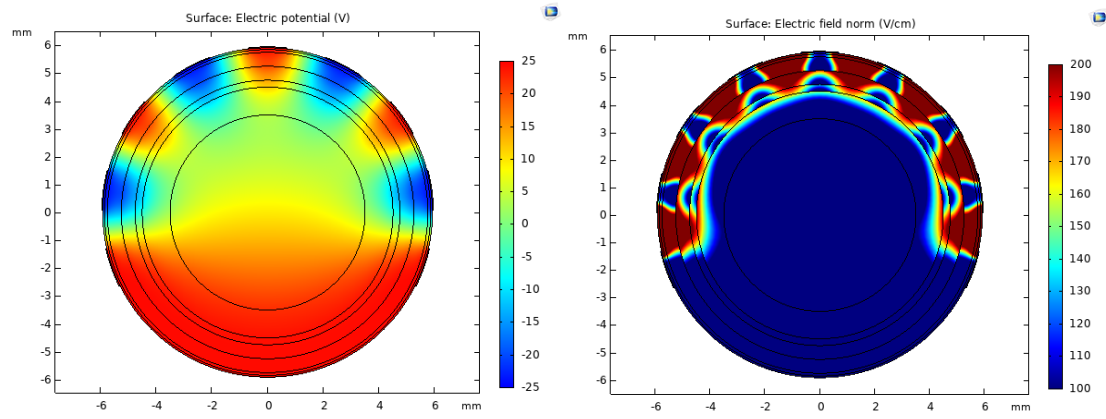


Figure 48: Electric potential distribution and Electric field distribution for advanced model

Chapter 6

6. Results and Discussion

The goal of this project was to design and optimize a helmet-type device for applying a high-magnitude electric field to the cortex of a rat's brain, within a voltage limit of 25V, for in-vivo applications related to Alzheimer's disease research. The simulations conducted using COMSOL Multiphysics provided valuable insights into the optimal electrode configurations required to achieve a uniform electric field with minimal voltage requirements.

Electric Field and Potential Distributions

The COMSOL simulations revealed that varying the width of the electrodes and the gaps between them significantly impacted the electric field and potential distributions on the cortex surface. Among the different width/gap ratios tested, the ratio of 1 (i.e., equal widths of electrodes and gaps) emerged as the most effective configuration.

1. **Electric Field Distribution:** The electric field norm images showed that the highest field strengths were concentrated near the electrodes. As the distance from the electrodes increased, the field strength decreased. Importantly, in the innermost circle of the setup, the electric field was almost zero, which is advantageous for protecting delicate tissues in the central region of the brain from excessive stimulation.
2. **Electric Potential Distribution:** The electric potential images displayed a smooth gradient from high potential near the electrodes to lower potential towards the center. This gradient ensured a stable and predictable electric field distribution, crucial for reliable in-vivo experiments. The optimized electrode configuration maintained symmetry and uniformity, which is essential for effective therapeutic application.

Optimization Analysis

The iterative simulations involved varying the width/gap ratios and analyzing parameters such as the depth of the electric field, maximum and minimum electric fields on the cortex, field uniformity, and the voltage required to achieve the target electric field strength of 150 V/cm. The results indicated that the width/gap ratio of 1 provided the best balance between these parameters.

1. **Depth and Field Strength:** The depth at which the electric field of 150 V/cm occurred was maximized at a width/gap ratio of 1, indicating effective penetration of the field into the cortex. Both the maximum and minimum electric field strengths on the cortex were highest at this ratio, ensuring a robust field application.
2. **Uniformity and Voltage Efficiency:** The uniformity of the electric field, measured as the difference between the maximum and minimum field strengths, was optimal at the width/gap ratio of 1. This configuration also required the least voltage to achieve the desired field strength, enhancing the efficiency and practicality of the setup.
3. **Number of Electrodes:** The analysis showed that the number of electrodes increased with the width/gap ratio. However, the ratio of 1 provided an optimal trade-off between the number of electrodes and the uniformity and strength of the electric field, making it the most efficient design.

3D Printing and Advanced Model

To further understand the spatial dimensions and anatomical relationships, 3D models of the mouse brain and skull were printed using a Formlabs SLA 3D printer. These models provided a tangible representation of the anatomical features, aiding in the design and optimization of the electrode setup. The advanced model included additional elements such as the scalp layer and a polyimide layer for fabricating the flexible printed circuit board (PCB). This comprehensive approach ensured the design's suitability for real-world applications.

The project successfully demonstrated that a width/gap ratio of 1 is the optimal configuration for achieving a uniform electric field and potential distribution on the rat cortex. This configuration ensures efficient voltage application, effective field penetration, and protection of delicate tissues. The integration of 3D printed models further enhanced the understanding and optimization of the setup. The results provide a promising avenue for non-invasive treatment of neurological conditions such as Alzheimer's disease, paving the way for future experimental validation and in-vivo applications.

Chapter 7

7. Future Scope

- AC analysis of 2D model mentioned in this report.
- Simulation of optimized 2D model of mouse head into 3D model.
- Printed Circuit Board (PCB) design and integration of power supply with electrodes
- Fabrication of device intended to be used on rat models having Alzheimer's disease

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